# A Tale of Two Chinas: Trade, Spatial Reallocation, and Uneven Structural Change

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**Abstract**

China's integration into world markets fueled spectacular growth in its manufacturing regions. This paper shows that such growth came at the expense of *another China*: regions outside the manufacturing core. Exploiting reduced U.S. trade policy uncertainty following China's WTO accession, I find that greater exposure expanded manufacturing in core counties but reduced it in the periphery. While core counties diversified, peripheral counties failed to absorb labor released from agriculture and experienced industrial downgrading. A spatial general equilibrium model with heterogeneous regional productivity distributions and internal migration accounts for these patterns, showing that trade integration can exacerbate pre-existing spatial inequality.

**Keywords:** Trade policy uncertainty; Spatial inequality; Structural change.

**JEL Codes:** R12, F13, F14, O14.

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# 1 Introduction

How does improved access to world markets reshape the spatial distribution of economic activity and the process of structural change within a country? In export-oriented economies, regions with greater exposure to trade shocks typically reap larger gains and industrialize more rapidly (McCaig and Pavcnik, 2018; Erten and Leight, 2021). China's post-WTO trajectory is often viewed as the canonical example of this dynamic: rapid manufacturing growth concentrated in coastal provinces appears to validate the link between trade exposure and local prosperity. However, this narrative implicitly assumes that the effects of greater trade exposure run in the same direction across regions. This paper challenges that assumption. Specifically, I ask: even in a quintessential export-oriented economy such as China, does greater exposure to improved access to foreign markets invariably generate stronger local growth and deeper industrialization?

The evidence suggests otherwise. In this paper, I reveal a trade-induced relative decline within *another China*: in regions lying outside the country's manufacturing heartland, more-exposed counties fared worse than their less-exposed peers. Exploiting the sharp reduction in trade policy uncertainty (TPU) triggered by the U.S. granting of Permanent Normal Trade Relations (PNTR) to China, I find sharply divergent local responses to the same trade shock. Specifically, reduced uncertainty sparked relative industrial expansion in the manufacturing core. In the periphery, by contrast, greater exposure produced relative hollowing out, including a shift toward lower-complexity manufacturing industries and weaker job creation. Although the trade shock moved labor out of agriculture in more-exposed periphery counties, it did not produce equal growth in local manufacturing employment. Instead, adjustment occurred through migration networks: only in core counties was employment growth sufficient to offset the pull

of historically connected migration destinations. These findings suggest that local responses to improved foreign market access reflect the interaction between pre-existing productivity differences in the tradable sector and labor mobility.

In 2000, the U.S. granted China PNTR status, eliminating the risk of future tariff escalation associated with annual renewals and securing permanent access to most-favored-nation (MFN) rates for Chinese exports. Although applied tariffs remained unchanged, the resulting reduction in TPU triggered significant firm responses. Theoretically, TPU acts as an investment wedge, forcing firms to delay costly market entry and upgrading decisions (Handley and Limão, 2022). To quantify local exposure to this shock, I construct a county-level shift-share measure—known as the "NTR gap"—by combining industry-level differences between MFN and non-MFN tariffs with the pre-treatment composition of local manufacturing employment (Pierce and Schott, 2016). Higher NTR gaps indicate greater exposure to the policy-induced reduction in TPU. Then, I link the NTR gap to a series of manufacturing outcomes using a standard difference-in-differences (DID) framework. Since non-MFN tariffs are largely inherited from the Smoot-Hawley Tariff Act of 1930 and are plausibly exogenous to contemporary local economic conditions, this approach allows me to estimate the causal effect of differential exposure to TPU reduction on relative changes in local manufacturing activity. Finally, to isolate the effect of TPU reduction, I control for concurrent changes in China's import tariffs and the tariffs faced by Chinese exports in major destination markets, while allowing for flexible time trends based on counties' administrative status, coastal location, and baseline employment share of state-owned enterprises (SOEs).

Consistent with previous studies, I find a positive average exposure gradient in local manufacturing activity: counties with higher NTR gaps experienced larger increases in firm count, employment, sales, and exports. A one-standard-deviation increase in the NTR gap raised these

outcomes by 5.3 to 17.8 percent relative to less-exposed counties. These average gains, however, mask stark spatial heterogeneity. To examine this heterogeneity, I stratify counties by their pre-treatment manufacturing employment share, defining the top quartile as the "core" and all remaining counties as the "periphery".[1] This distinction captures differences in productivity in the tradable sector: regions with established industrial bases are better positioned to capitalize on export opportunities, whereas less-industrialized areas have fewer productive firms capable of doing so. Because regions are linked through labor mobility, industrial expansion in the core can induce workers to leave the periphery. This framework implies divergent responses to the same policy shock: higher exposure is expected to accelerate manufacturing expansion within core counties, but may dampen manufacturing growth within periphery counties relative to their less-exposed peers, leading to a relative decline in the periphery's manufacturing position.

The empirical evidence confirms this divergence. Within the core, counties with greater TPU reductions experienced sizable increases in key manufacturing outcomes. For example, a one-standard-deviation higher NTR gap increased employment by 9.4 percent and exports by 24.5 percent among core counties. In contrast, within the periphery, the same increase in exposure reduced employment by 3.6 percent and exports by 14.8 percent. A pooled specification interacting exposure with core status confirms that these gradients differ statistically: the interaction is positive and significant across all four manufacturing outcomes. Thus, greater exposure generated opposite-signed relative responses across space, inducing manufacturing

[1]The terms "core" and "periphery" are drawn from the dependency theory (Prebisch, 1950; Wallerstein, 1974) and new economic geography (Krugman, 1991; Fujita et al., 2001) traditions. I use them to denote counties with different initial levels of industrialization rather than by geography, classified by their pre-treatment manufacturing employment shares. The results do not depend on the quartile cutoff: Section 5.6 shows that the core–periphery contrast persists when the core is defined as the top sixth, eighth, or tenth of the distribution.

expansion within the core but contraction within the periphery. These patterns are consistent with TPU reduction acting as a catalyst for spatial reallocation.

These divergent responses operate primarily through extensive-margin adjustment rather than changes among incumbent firms. A decomposition of the estimated effects shows that, within the core, greater exposure increased the number of observed domestic firms and exporters; within the periphery, greater exposure instead reduced the number of exporters. In contrast, I find little evidence of differential adjustment along the intensive margin: incumbent firms show negligible changes in employment or sales. These findings are consistent with trade models incorporating firm heterogeneity (Melitz, 2003; Chaney, 2008), in which trade shocks generate adjustment through selection, including changes in firm participation and export status.

Finally, I document that differential exposure to the TPU reduction induced uneven structural change, both within manufacturing and across the broader sectoral composition of local economies. Within manufacturing, more-exposed periphery counties experienced industrial downgrading: employment shifted out of machinery and light manufacturing and into food processing, lowering the county-level complexity index. By contrast, greater exposure expanded industrial participation in the core, with new entrants emerging in both upstream and downstream industries, thereby preserving the region's production complexity. These exposure gradients also extended beyond manufacturing. Using population census data, I find that although agricultural employment declined nationwide over this period, the pattern of structural adjustment differed across regions. In more-exposed periphery counties, declining agricultural employment was not offset by manufacturing expansion, revealing the periphery's limited capacity for structural transformation.

A battery of identification and robustness checks supports the causal interpretation. I find no differential pre-trends, and inference remains robust to alternative assumptions about the underlying shift-share shocks, specifications, outlier exclusions, and placebo tests. Moreover, finer partitions of initial industrialization show that positive exposure gradients are always concentrated among the most industrialized counties.

To understand the mechanisms underlying this divergence, I first examine why regions differ in their capacity to respond to improved foreign market access. A key source of heterogeneity lies in the distribution of firm productivity. Estimating region-specific Pareto shape parameters from the upper tail of the firm size distribution, I find significantly thicker right tails in the core among both incumbent and entering firms. The greater mass of highly productive firms in core counties allows them to respond more strongly to expanded export opportunities, consistent with the extensive-margin adjustment documented above. However, differences in productive capacity alone do not explain why greater exposure reduces manufacturing employment in the periphery. Because regions are connected through labor mobility, stronger industrial expansion in the core can draw workers away from the periphery. Consistent with this mechanism, a core county's own exposure to TPU reduction improves its net migration balance, whereas a periphery county's own exposure induces no comparable response. Instead, periphery counties historically connected to more-exposed migration destinations experience deteriorating migration balances and weaker manufacturing employment.

I develop a spatial general equilibrium model with heterogeneous firms and internal migration to formalize the mechanisms above. The model incorporates regional differences in the thickness of Pareto productivity tails, which govern how strongly manufacturing responds to improved foreign market access. A reduction in trade costs generates two opposing forces in

the periphery. First, a *wage effect* draws workers from agriculture into manufacturing in both regions, with a stronger response in the core because of its thicker productivity tail. Second, the resulting asymmetry in manufacturing expansion induces a *migration effect*, pulling workers from the periphery toward the core. The model shows that when labor is sufficiently mobile, this migration effect can dominate the local wage effect in the periphery, causing manufacturing employment to decline despite rising wages. Thus, even a favorable trade shock can produce opposite-signed regional employment responses without imposing productivity losses or adverse local trade shocks on the periphery.

Taken together, this paper provides novel evidence on the spatially uneven effects of China's integration into world markets. In response to a common reduction in TPU, greater exposure induced manufacturing expansion within the core, while more-exposed counties in the agrarian periphery experienced firm exit, industrial downgrading, and stagnation relative to their less-exposed peers. Crucially, initial productivity heterogeneity determined where expanded market access translated into local expansion, while internal migration amplified the resulting spatial reallocation by drawing workers toward the more productive core. In effect, high pre-PNTR uncertainty functioned as implicit protection for manufacturing activity in lagging regions, while its removal reinforced existing productivity advantages. Consequently, rather than fostering regional convergence, deeper trade integration can entrench spatial disparities in industrial activity by cementing the dominance of the manufacturing core. This contrast—between a core propelled toward further industrialization and globalization and a periphery pushed toward relative industrial decline—is, ultimately, a tale of two Chinas.

**Related Literature.** This paper contributes to three strands of literature. First, it builds on the literature on the regional incidence of international trade shocks. Existing work shows how initial industrial specialization translates industry-level trade shocks into uneven local labor-market outcomes (Autor et al., 2013; Pierce and Schott, 2016; Acemoglu et al., 2016; Kovak, 2013; Dix-Carneiro and Kovak, 2017, 2019), while studies of developing countries document structural adjustment following improved export access (McCaig and Pavcnik, 2018). Most closely, Erten and Leight (2021) find that greater exposure to China's PNTR-induced TPU reduction accelerated structural transformation on average. Recent work has uncovered heterogeneity behind such aggregate effects: Lin et al. (2024) emphasize initial agricultural endowments, while He and Qi (2024) attribute spatial divergence to entrepreneur relocation between coastal and inland areas. Within rural China, Ma et al. (2025) find that higher PNTR exposure induces selective exit from agriculture and lowers agricultural productivity. I find that the average effect of PNTR exposure masks opposite gradients across initial industrial conditions: greater exposure expands manufacturing in the industrial core but induces relative contraction in the periphery. This divergence reflects differences in pre-existing manufacturing capacity and operates through firm selection. Industrial structure therefore shapes not only local exposure to trade shocks but also whether such exposure translates into industrial expansion.

Second, this paper contributes to the literature on trade policy uncertainty. Existing work shows that TPU delays firm entry, investment, and export decisions (Handley, 2014; Handley and Limão, 2015, 2017, 2022), while studies exploiting China's PNTR accession document effects on exports, firms, and internal migration (Feng et al., 2017; Facchini et al., 2019; Cui and Li, 2023; Yuan and Ouyang, 2025). Departing from the focus on average responses, I show that TPU reduction reallocates economic activity across locations, as differences in tradable-sector

productivity interact with internal migration. It can therefore reinforce, rather than attenuate, pre-existing spatial disparities within an integrating economy.

Finally, this paper connects heterogeneous-firm trade theory with structural transformation and economic geography. Models with firm heterogeneity emphasize adjustment through entry, exit, and selection (Melitz, 2003; Chaney, 2008; Demidova and Rodriguez-Clare, 2013), while theories of uneven structural transformation show how integration can generate divergent outcomes across regions (Matsuyama, 2009; Kaboski et al., 2026). Quantitative work on China further highlights how internal trade and migration frictions shape the spatial consequences of external integration (Coşar and Fajgelbaum, 2016; Tombe and Zhu, 2019; Fan, 2019), while Tian (2024) shows that trade integration itself can relax such frictions through more migrant-friendly local *hukou* regulations. I bring firm heterogeneity into this perspective by linking regional responses to differences in the upper tail of firm productivity distributions, following Bakker et al. (2024), while showing how internal migration propagates adjustment across space. Initial manufacturing specialization, rather than population agglomeration *per se*, better accounts for this regional heterogeneity. In the presence of such heterogeneity, greater exposure to a favorable trade shock can accelerate relative decline where initial industrial capacity is weak.

The remainder of the paper is organized as follows. Section 2 outlines the policy background of the PNTR status and the institutional context of China's WTO accession. Section 3 describes the data sources and sample construction. Section 4 details the identification strategy. Section 5 presents the main results and robustness checks. Section 6 examines the mechanisms underlying the spatial divergence, providing evidence on regional productivity differences and internal migration. Section 7 develops a theoretical framework that rationalizes these patterns within a spatial general equilibrium model with heterogeneous firms. Section 8 concludes.

# 2 Background

In 2000, the U.S. Congress passed legislation granting China Permanent Normal Trade Relations (PNTR) status contingent upon its WTO accession, ending decades of policy uncertainty. Prior to this, China's access to the U.S. market was contingent on an annual review process under the Trade Act of 1974. To maintain low Most-Favored-Nation (MFN) tariff rates, China required a yearly presidential waiver; without it, tariffs would revert to the prohibitive "Column 2" rates established by the Smoot-Hawley Tariff Act of 1930, which were substantially higher than MFN rates for nearly all product categories. Although China had effectively enjoyed MFN status since 1980, the annual renewal became increasingly contentious following the political turmoil in 1989. Congress repeatedly threatened to revoke the waiver over human rights and geopolitical concerns, creating uncertainty where low-tariff access to the U.S. market was treated as a temporary privilege rather than a permanent right.

The annual waiver requirement stood in direct conflict with the WTO's core principle of unconditional most-favored-nation treatment. As China approached the final stages of accession, this incompatibility threatened to trigger the WTO's non-application clause, which would have effectively denied U.S. firms the benefits of China's market liberalization commitments. To resolve this dilemma, Congress enacted legislation granting China PNTR status in mid-2000, contingent on the country's WTO admission. President Clinton signed the act into law on October 10, 2000, marking the end of the annual review era. PNTR status officially took effect on December 27, 2001, shortly after China's formal WTO accession on December 11.

The granting of PNTR status permanently eliminated the threat of tariff escalation, thereby resolving the trade policy uncertainty that had historically inhibited investment. Following this

policy change, China's exports to the U.S. surged, nearly quadrupling from $102.3 billion in 2001 to $399.4 billion in 2011. This expansion was primarily driven by the manufacturing sector, which had been the main target of potential protective tariffs. From 2000 to 2013, machinery, textiles, and apparel together accounted for an average of 61.3 percent of China's total exports, followed by metals, transport equipment, and chemical products (Brandt and Lim, 2024). Given that the reduction in TPU predominantly affected the manufacturing sector, my empirical analysis focuses on this segment of the economy.

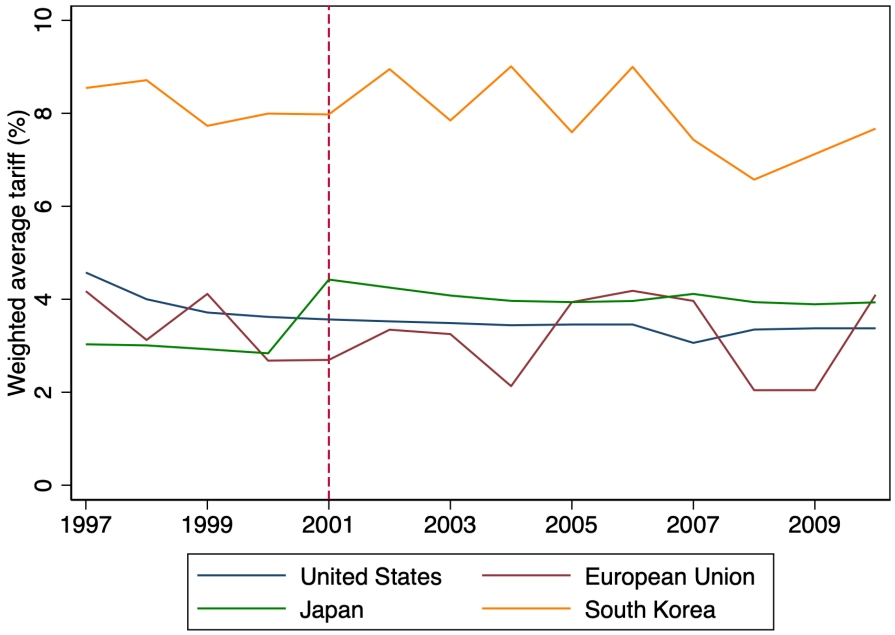



Figure 1: Applied tariff rates on Chinese exports, 1997-2010

*Notes:* This figure plots the applied tariff rates on Chinese exports in their four largest destination markets. Tariff data are sourced from the WITS database.

To isolate the causal impact of the U.S. PNTR agreement, it is crucial to establish that this policy shock was not confounded by concurrent changes in applied tariff rates or policy shifts in other major export markets surrounding China's WTO accession. In the U.S., Chinese exporters had already benefited from low MFN rates since 1980, meaning PNTR altered the certainty of

market access rather than the tariff level itself. Crucially, this policy stability extended to China's other major trading partners, ensuring that the reduction in uncertainty was unique to the U.S.–China relationship. Prior to its accession to the WTO, China had already institutionalized commercial relations with Europe and Japan through bilateral trade agreements signed in the 1970s and 1980s, leaving South Korea as the only top export destination without a formal agreement.[2] Aside from PNTR, no major diplomatic milestones significantly restructured China's external trade environment during this period. This stability is corroborated by Figure 1, which plots the weighted average applied tariff rates of partners accounting for 81.8 percent of China's 2001 exports (including Hong Kong, a free port). With the exception of South Korea, these rates remained low and exhibited no trend breaks around 2001, confirming that the reduction in TPU was a shock specific to the U.S. market.

# 3 Data

## 3.1 Administrative Divisions of China

I use the county as the fundamental geographic unit of analysis. As the third tier of China's administrative hierarchy, the county represents the lowest level of government with substantial administrative autonomy and consistent statistical reporting. A significant challenge in longitudinal studies of China is the frequent restructuring of these administrative divisions, particularly the reclassification of counties into districts. To address this, I construct a time-consistent concordance that harmonizes administrative boundaries across the sample period. This procedure

[2]These agreements include the EEC-China Trade Agreement (1978), the EEC-China Trade and Cooperation Agreement (1985), the Japan-China Trade Agreement (1974), and the Japan-China Treaty of Peace and Friendship (1978). The China-South Korea Free Trade Agreement, signed in 2015, falls outside my sample period.

creates a unified panel encompassing both rural counties and urban districts, allowing me to capture granular spatial heterogeneity often obscured in prefecture-level studies. Unlike approaches that exclude urban cores to avoid boundary complications, my framework explicitly incorporates them, enabling a direct comparison between urban districts and their surrounding counties. I detail the construction procedure and provide illustrative examples in Appendix C. The final dataset comprises 2,268 consistent county-level units (hereafter "counties"), including 292 merged urban districts.

## 3.2 Manufacturing Outcomes

Data on local manufacturing activity are constructed from firm-level microdata provided by the Annual Survey of Industrial Firms (ASIF) conducted by the National Bureau of Statistics (NBS). Spanning the period from 1998 to 2007, this survey covers all state-owned enterprises (SOEs) and non-SOEs with annual sales exceeding 5 million RMB. To ensure sample consistency, I restrict the analysis to the manufacturing sector and apply the revenue threshold to all firms. Following standard protocols, I clean the data by removing observations with negative or missing key values and excluding firms with fewer than eight employees to mitigate measurement error (Brandt et al., 2012; Nie et al., 2012). The final sample, which expanded from approximately 110,000 firms in 1998 to 302,000 in 2007, is spatially linked to my time-consistent county-level administrative units using a hierarchical matching algorithm based on administrative division codes, postal codes, and textual addresses. I then aggregate these firm-level observations to calculate county-level total number of firms, employment, sales, and export value, with all

monetary variables deflated to 1998 constant prices using the World Bank deflator series for China. Panel A of Table A1 presents summary statistics for these aggregate outcomes.

A potential limitation of the ASIF is its restriction to "above-scale" firms, which may omit the dynamics of smaller enterprises. However, this threshold-induced selection bias is likely minimal for the county-level outcomes of interest. Comparisons with the 2004 First National Economic Census reveal that ASIF-covered firms capture the vast majority of manufacturing activity, accounting for 71.2 percent of total employment, 90.7 percent of output, and 97.5 percent of exports (Brandt et al., 2014). The county-level average export intensity derived from the ASIF is 9.26 percent, closely tracking the Census benchmark of 8.59 percent (Bakker et al., 2024). These statistics confirm that the dataset provides a representative coverage of China's local manufacturing capacity, particularly for export-oriented activities.

A second challenge is the prevalence of zero values in the aggregate data. These should not be interpreted as missing data: all counties are covered by the ASIF, and zeros arise when no manufacturing firm above the reporting threshold is observed in a county-year. While zeros in total manufacturing employment or sales are relatively rare, they are prevalent in exports, which are heavily concentrated in more developed counties. For example, 1,028 counties (nearly half the sample) reported zero manufacturing exports in 1998, and over 500 counties switched exporting status during the sample period. These zeros capture an important extensive margin of adjustment. Excluding them would produce an unbalanced panel and risk sample selection by omitting precisely the entry and exit dynamics that may drive spatial divergence. I therefore retain zero observations and use the Poisson pseudo-maximum likelihood (PPML) estimator as my baseline specification in Section 4, which naturally accommodates them. The base year is the one exception: 231 counties had no above-scale manufacturing firm in 1998, so the

employment weights in equation (2) are undefined and no NTR gap can be constructed. These counties are therefore excluded, leaving 2,037 counties in the estimation sample.

### 3.3 Trade Policy Uncertainty

Following the literature, I construct a Bartik-like shift-share measure to quantify the differential exposure of Chinese counties to TPU reduction in the U.S. market (Bartik, 1991; Pierce and Schott, 2016; Erten and Leight, 2021). The measure combines an industry-specific "shift"—the magnitude of the potential tariff hike—with a local "share"—the county's initial industrial composition. Specifically, the shift component is the industry-level NTR gap, defined as the spread between the prohibitive non-NTR ("Column 2") tariff rate and the applied MFN rate. The share component is the county's manufacturing employment structure prior to the uncertainty shock. By fixing the industrial composition before the shock, the measure isolates the impact of the external policy change. Formally, for county $c$ and industry $i$, I define the industry-level gap (NTR Gap$_i$) and the county-level exposure (NTR Gap$_c$) as:

$$\text{NTR Gap}_i = \text{Non-NTR Rate}_i - \text{NTR Rate}_i, \tag{1}$$

$$\text{NTR Gap}_c = \sum_i \frac{L_{ci,1998}}{L_{c,1998}} \times \text{NTR Gap}_i, \tag{2}$$

where $L_{ci,1998}$ denotes employment in industry $i$ in county $c$ in 1998, and $L_{c,1998}$ represents the county's total manufacturing employment in that year.[3] The term in parentheses represents the share of industry $i$ in the county's total manufacturing employment. Since the granting of PNTR

[3]The employment shares are defined over manufacturing industries and sum to one within each county by construction, placing the design in the complete-shares case of Borusyak et al. (2022). Section 5.5 explains why excluding non-manufacturing industries matters for identification and reports the corresponding diagnostics.

status permanently eliminated the threat of tariff escalation, a higher NTR Gap$_c$ corresponds to a greater reduction in trade policy uncertainty.

Constructing this measure requires addressing the classification mismatch between U.S. tariff codes (defined at the HS 8-digit level) and Chinese firm employment (recorded at the CIC 4-digit level). I obtain the raw tariff data from Pierce and Schott (2016) and calculate $G_c$ using a three-step procedure. First, I aggregate the 8-digit tariff gaps to the HS 6-digit level, weighting by total U.S. import values in 1998.[4] Second, I map these HS 6-digit gaps to the CIC 4-digit level using the concordance provided by Brandt et al. (2017), which harmonizes international trade codes with domestic industry classifications while adjusting for the 2002 CIC system revision.[5] Finally, I combine the matched industry gaps ($G_i$) with county-level employment weights ($L_{ci}$) to compute the final exposure measure following equation (2).

Figure A1 illustrates the distribution of NTR gaps across manufacturing industries and counties. The figure shows substantial heterogeneity: industry-level gaps range from zero to 86.3 percent, while county-level gaps run from zero (in four counties) to 64.1 percent. Table A2 lists the highest- and lowest-gap industries among those that account for at least 0.1 percent of manufacturing employment. The highest NTR gaps are concentrated in labor-intensive sectors, such as toys, plastics, and apparel, whereas lower gaps appear in industries related to agro-processing, fertilizers, and raw materials. This cross-industry variation in NTR gaps aligns with previous calculations using U.S. industrial classifications (Handley and Limão, 2022) and is inherited by counties specialized in the corresponding production activities.

[4] An alternative weighting method uses U.S. import values *from China* in 1998, yielding similar results. However, this approach mechanically assigns a zero gap to industries with no initial Chinese exports, potentially ignoring sectors where trade was inhibited precisely due to high uncertainty. Using total U.S. import values avoids this bias. U.S. import data at the 10-digit HS level are sourced from Schott (2008).

[5] The CIC system underwent a structural update in 2002. The concordance of Brandt et al. (2017) addresses this by providing a harmonized list of industry codes consistent before and after the change.

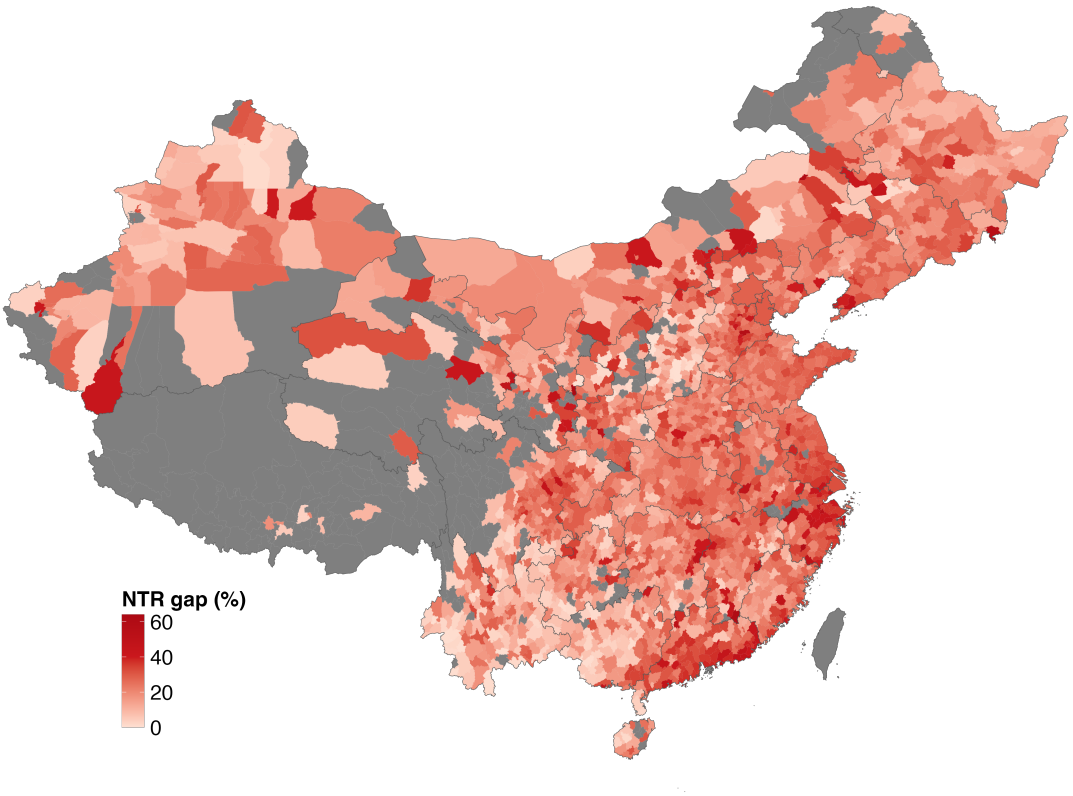



Figure 2: Geographic Distribution of County-level NTR Gaps

*Notes:* This figure illustrates the geographic distribution of NTR gaps across Chinese counties. County-level NTR gaps are constructed using a shift–share measure that combines industry-level NTR gaps with counties' manufacturing employment composition in 1998. Counties outside the estimation sample because an NTR gap cannot be constructed are shaded in gray.

Figure 2 maps the geographic distribution of these shocks at the county level. The average county-level NTR gap is 21.61 percent, indicating that Chinese manufacturing as a whole faced significant uncertainty in U.S. markets prior to PNTR. However, the map reveals substantial spatial variation driven by local industrial endowments. High-exposure counties are not strictly confined to specific macro-regions but are scattered across areas with historical concentrations of labor-intensive manufacturing, such as textiles and light assembly. Crucially, this variation exists even within provinces and prefecture-level cities. For instance, neighboring counties can

exhibit drastically different exposure levels depending on whether they specialize in highly protected industries (e.g., toys) or low-gap ones (e.g., agro-processing). This granular spatial heterogeneity, which stems from the pre-existing industrial composition, provides the variation necessary to identify the causal effects of TPU reduction. Counties with no reported manufacturing employment in 1998 are excluded from the sample, as their NTR gaps are mathematically undefined. Given that these counties contribute negligibly to aggregate manufacturing output, their exclusion is unlikely to bias the results.

## 3.4 Other Control Variables

To isolate the causal impact of TPU reduction from concurrent changes in trade costs, I explicitly control for applied tariff levels. Specifically, I construct county-level measures of tariffs imposed on Chinese exports by its four largest trading partners: the United States, the European Union, Japan, and South Korea. Additionally, I control for China's import tariffs to capture the effects of import competition and access to intermediate inputs. Tariff data are obtained from the WITS database and aggregated to the county level following the shift-share approach in equation (2). The data reveal a clear divergence in tariff dynamics: while China's average import tariff rate declined substantially from 17.4 percent in 1998 to 9.0 percent in 2007, the applied tariffs facing Chinese exporters remained historically low and stable. Notably, the average U.S. applied tariff dropped only marginally from 2.8 percent to 2.4 percent, confirming that the primary shock was the removal of uncertainty rather than a reduction in tariff levels.

I also account for the confounding influence of pre-existing economic structure and geographic endowments. First, regarding domestic State-Owned Enterprise (SOE) restructuring, I

use the 1998 ASIF data to calculate each county's initial SOE share, classifying firms as SOEs if state ownership exceeds 50 percent of paid-in capital. Controlling for this is crucial because the massive retrenchment of the state sector released a significant amount of labor that was subsequently absorbed by the private manufacturing sector. Finally, to account for geographic characteristics that determine market access, I calculate the minimum distance from each county's centroid to the nearest port (defined as either a coastline or a navigable river) using GIS data. To flexibly capture the differential secular trends driven by these initial conditions, I control for linear time trends interacted with quartiles of the initial SOE share and quartiles of the distance to the nearest port.

# 4 Empirical Strategy

## 4.1 Model and Identification

I estimate the causal effects of TPU reduction on county-level manufacturing outcomes using a constant-elasticity difference-in-differences (DID) specification with two-way fixed effects. Specifically, I assume that the expected value of the outcome variable $Y_{ct}$ in county $c$ and year $t$ takes an exponential form:

$$\mathbb{E}[Y_{ct}] = \exp\left[\beta \cdot (\text{NTR Gap}_c \times \text{Post}_t) + \delta_c + \lambda_t + \mathbf{X}'_{ct}\boldsymbol{\gamma} + \mathbf{Z}'_c t\right], \tag{3}$$

where $\text{NTR Gap}_c$ is the county-level measure of TPU exposure defined in equation (2), normalized to mean zero and standard deviation one to facilitate interpretation. $\text{Post}_t$ is an indicator variable taking the value of one for years 2001–2007 and zero otherwise. $\delta_c$ and $\lambda_t$ denote

county and year fixed effects, respectively. The vector $\mathbf{X}_{ct}$ contains time-varying tariff controls. $\mathbf{Z}_c$ captures predetermined county characteristics, including province, administrative status, and quartiles of both the initial SOE share and the distance to the nearest port. These characteristics are interacted with a linear time trend ($\mathbf{t}$) to control for differential secular trends driven by initial economic structure and geographic endowments.

The coefficient of interest, $\beta$, captures the semi-elasticity of manufacturing outcomes with respect to a one-standard-deviation reduction in TPU. Identification of $\beta$ relies on the parallel trends assumption: conditional on fixed effects and time-varying controls, counties with varying exposure to the TPU shock would have followed similar trajectories in the absence of the policy change. The validity of this assumption is supported by the exogeneity of the "shift" component in my exposure measure (Borusyak et al., 2022). As discussed, the NTR gaps stem from the non-NTR tariff rates largely set by the Smoot-Hawley Tariff Act of 1930. These historical tariffs, established to protect U.S. industries during the Great Depression, are plausibly uncorrelated with the unobserved potential outcomes of Chinese counties in the late 1990s. Consequently, high-exposure counties should not systematically differ from low-exposure counties in terms of their pre-accession growth dynamics. In Section 5.4, I formally test this implication using an event study design and find no evidence of differential pre-trends.

## 4.2 Estimation Methods

I estimate $\beta$ using the Poisson Pseudo Maximum Likelihood (PPML) estimator. Developed by Santos Silva and Tenreyro (2006), PPML is widely used to estimate constant-elasticity models with non-negative dependent variables. The regression equation is

$$Y_{ct} = \exp\left[\beta \cdot (\text{NTR Gap}_c \times \text{Post}_t) + \delta_c + \lambda_t + \mathbf{X}'_{ct}\boldsymbol{\gamma} + \mathbf{Z}'_c t\right] + \varepsilon_{ct}, \tag{4}$$

where $Y_{ct} \geq 0$ and $\varepsilon_{ct}$ is clustered at the county level.

I use PPML instead of the standard log-linear OLS estimator for two primary reasons. First, as detailed in Section 3, the use of ASIF data results in zero values for some county-year observations, particularly for exports. This renders the logarithmic transformation infeasible without arbitrary adjustments. Common *ad hoc* solutions, such as the $\log(1+Y)$ transformation, are problematic because they lack scale invariance: changing the units of measurement can arbitrarily alter the estimated coefficients (Chen and Roth, 2024). On the other hand, PPML avoids this issue by modeling the dependent variable in levels, naturally accommodating zero outcomes without requiring arbitrary transformations.

Second, the economic size of counties in my sample varies substantially. In 1998, counties in the top quartile accounted for 85 percent of the total number of firms and employment, 90 percent of total sales, and 96 percent of total exports. This imbalance matters because PPML and log-linear OLS summarize heterogeneous local responses differently. Log-linear OLS gives similar weight to proportional changes in counties of very different economic size, whereas PPML places greater weight on counties with larger predicted outcomes. When responses are heterogeneous, log-linear OLS therefore approximates an average of county-specific semi-elasticities, while PPML approximates the semi-elasticity of the aggregate outcome (Tyazhelnikov et al., 2026). Moreover, log-linear OLS models the conditional mean of log outcomes rather than the log of the conditional mean outcome. Under heteroskedasticity, these two objects need not coincide

because of Jensen's inequality, potentially producing substantial differences between log-OLS and PPML estimates (McConnell, 2024).

My object of interest is the semi-elasticity of the conditional mean outcome with respect to exposure to TPU reduction. Under the exponential specification, a one-standard-deviation increase in exposure changes the conditional mean of a county-level manufacturing outcome by $\exp(\hat{\beta}) - 1$. This effect should be interpreted as a relative exposure gradient: since year fixed effects absorb changes common to all counties, the specification identifies neither the national aggregate effect of PNTR nor its absolute effect on counties at any particular level of exposure. I examine the robustness of the results to alternative estimation methods in Section 5.6.

## 4.3 Core and Periphery Counties

Chinese counties differ substantially in their stage of structural transformation, particularly in the extent of manufacturing development. As shown in Panel D of Table A1, the pre-treatment share of employment in manufacturing ranges from near zero to over 75 percent. This heterogeneity is central to the analysis because reductions in TPU operate as a selection mechanism. Although selection occurs at the firm level, its county-level consequences are inherently spatial, reflecting differences across counties in the distribution of firms relative to the export cutoff.

Standard trade theory predicts that trade liberalization disproportionately benefits high-productivity firms, which are better able to absorb fixed exporting costs. A county's response therefore depends on the prevalence of such firms. Throughout this paper, I take a more developed manufacturing base as a proxy for a thicker right tail of the local productivity distribution, a structural hypothesis examined directly in Section 6. Core counties are therefore

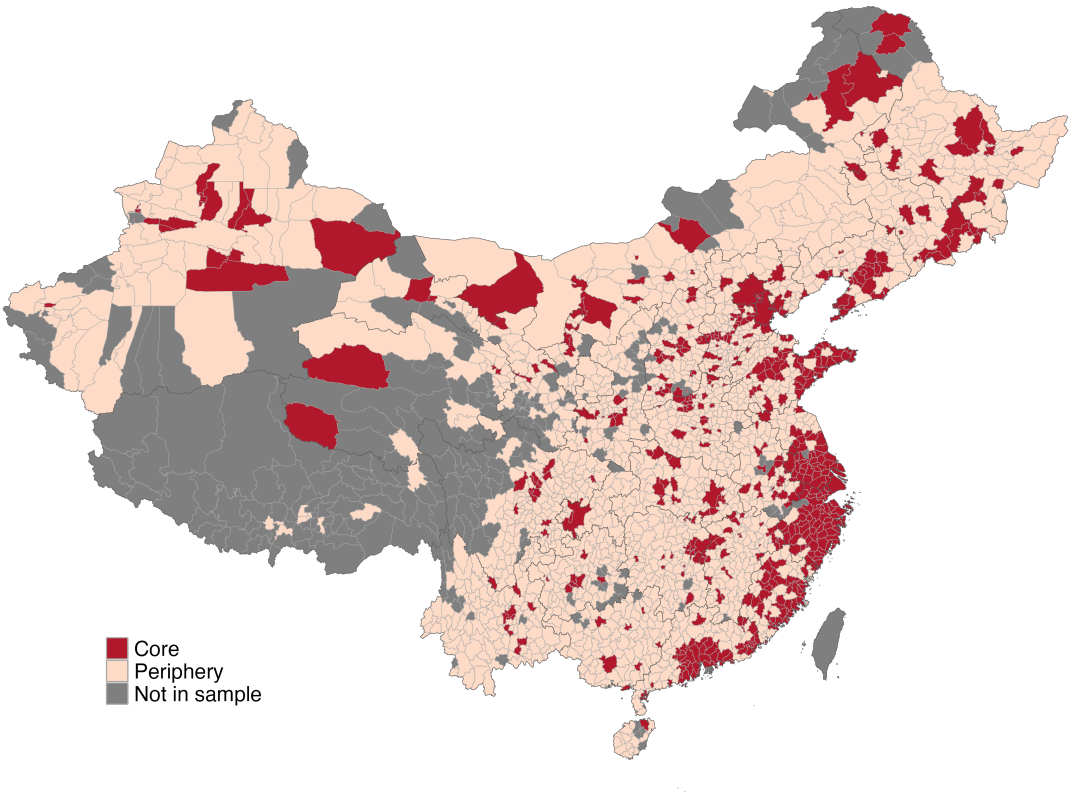



Figure 3: Geographic Distribution of Core-Periphery Division

*Notes:* This figure shows the classification of Chinese counties into core and periphery regions. Counties are ranked by their manufacturing employment share in 2000 and divided into quartiles; the core consists of counties in the top quartile, while the periphery consists of those in the bottom three quartiles. Counties outside the estimation sample because an NTR gap cannot be constructed are shaded in gray.

defined by more than industrial density: they hold a larger mass of firms close enough to the export cutoff to respond to the policy shock. Periphery counties exhibit more compressed productivity distributions and a thinner pool of globally competitive firms. In these areas, few firms sit close enough to the cutoff for improved market access to generate much entry. Because labor is mobile across regions, the expansion the shock produces in the core also raises the return to leaving the periphery, so exposure there operates mainly through the labor market. Guided by this logic, I classify counties by their pre-treatment manufacturing employment share:

those in the top quartile are designated as the core, while the remaining counties constitute the periphery. The model is then estimated separately for the two groups. In Section 5.6, I redefine core counties using alternative cutoffs and obtain similar results.

Figure 3 shows that this core–periphery division is correlated with the coast–inland gradient: both the Yangtze River Delta and the Pearl River Delta are classified as core. At the same time, the classification is more granular than a coastal indicator, identifying inland urban centers as core and agricultural areas surrounding major coastal clusters as periphery. Figure A2 compares pre-treatment economic structures in 2000. Periphery counties account for about 60 percent of the population and 62 percent of employment, yet only around 20 percent of manufacturing employment. Manufacturing firms in these counties are also smaller in terms of sales and, in particular, exports, which account for only about 5 percent of the national total.

# 5 Results

## 5.1 Average Effects

I first estimate the average effects of TPU reduction on county-level manufacturing outcomes, pooling all units under the assumption of homogeneous treatment effects. These results provide a benchmark for comparison with previous studies, such as Erten and Leight (2021).

Table 1 reports these estimates. Column (1) includes only two-way fixed effects and tariff controls in addition to the DID term. Columns (2) to (5) sequentially introduce interactions between county characteristics and time trends to account for potential confounding factors. Consistent with the existing literature, TPU reduction significantly increases all outcome vari-

Table 1: Average Effects of TPU Reduction

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A. Firms** | | | | | |
| NTR gap × Post | 0.111*** | 0.0532** | 0.0732*** | 0.0752*** | 0.0807*** |
| | (0.0230) | (0.0209) | (0.0229) | (0.0230) | (0.0233) |
| **Panel B. Employment** | | | | | |
| NTR gap × Post | 0.174*** | 0.0456** | 0.0697*** | 0.0661*** | 0.0533** |
| | (0.0331) | (0.0209) | (0.0213) | (0.0212) | (0.0223) |
| **Panel C. Sales** | | | | | |
| NTR gap × Post | 0.0409 | 0.0312 | 0.0535** | 0.0526** | 0.0517* |
| | (0.0275) | (0.0250) | (0.0262) | (0.0261) | (0.0275) |
| **Panel D. Exports** | | | | | |
| NTR gap × Post | 0.199*** | 0.159*** | 0.161*** | 0.165*** | 0.164*** |
| | (0.0457) | (0.0446) | (0.0437) | (0.0436) | (0.0438) |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | ✓ | ✓ | ✓ | ✓ | ✓ |
| Trends: Province | | ✓ | ✓ | ✓ | ✓ |
| Trends: Admin Status | | | ✓ | ✓ | ✓ |
| Trends: Port Quartile | | | | ✓ | ✓ |
| Trends: SOE Quartile | | | | | ✓ |
| # of Obs. (Panel A-C) | 20,370 | 20,370 | 20,370 | 20,370 | 20,370 |
| # of Obs. (Panel D) | 17,220 | 17,220 | 17,220 | 17,220 | 17,220 |

*Notes:* This table presents the average effects of TPU reduction on manufacturing outcomes, assuming homogeneous treatment effects across counties. Each cell corresponds to a separate regression, with the dependent variable indicated in the panel header. All specifications are estimated using PPML, with standard errors clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

ables. Under the baseline specification (Column 5), a one-standard-deviation increase in the NTR gap leads to an increase of 8.4 percent in the number of manufacturing firms, 5.5 percent in employment, and 5.3 percent in sales. The impact on exports is even more pronounced, with

growth reaching 17.8 percent.[6] These magnitudes indicate that, after China's WTO accession, counties more exposed to TPU reduction grew substantially faster than less-exposed counties, with effects comparable in scale to the estimates in Erten and Leight (2021).

## 5.2 Spatial Reallocation

I now turn to the spatial reallocation induced by the reduction in TPU. Table 2 shows that the exposure gradient reverses sign between core and periphery counties. Within the core, a one-standard-deviation higher NTR gap increases manufacturing employment by 9.4 percent and exports by 24.5 percent. Within the periphery, by contrast, it reduces these outcomes by 3.6 percent and 14.8 percent, respectively. The core and periphery gradients are statistically different for all four outcomes.[7] Since the two groups are linked through labor mobility, these opposing gradients are consistent with spatial reallocation: export expansion in the core draws workers away from periphery counties, limiting their capacity to expand manufacturing.

As set out in Section 4.2, these coefficients are exposure gradients. The negative estimates for periphery counties therefore do not imply an absolute decline in the periphery's manufacturing sector, nor do they contradict the positive average effects documented earlier, which are driven by the core's dominant economic size. Figure A3 makes both points visible in the raw data. Manufacturing grew in absolute terms in both groups and in both halves of the exposure

[6] PPML estimates can be converted to percentage changes by calculating $\left(e^{\hat{\beta}\cdot\Delta X}-1\right)\times 100$, where $\hat{\beta}$ is the estimated coefficient from the PPML regression and $\Delta X$ represents NTR gap changes in standard deviations.

[7] Tables A4 and A5 report separate estimates for core and periphery counties under each of the specifications reported in Table 1, while Table A6 reports the corresponding core–periphery differences. The core–periphery difference is significant at the one percent level for all four outcomes across all specifications. Table A7 confirms this qualitative pattern in a pooled specification that interacts exposure with core status while imposing common coefficients and trends across groups.

Table 2: Spatial Reallocative Effects of TPU Reduction

| | (1) Firms | (2) Employment | (3) Sales | (4) Exports |
|---|---|---|---|---|
| **Quartiles of Manufacturing Employment Share** | | | | |
| **Panel A. Core** | | | | |
| NTR gap × Post | 0.122*** | 0.0900*** | 0.0634* | 0.219*** |
| | (0.0359) | (0.0323) | (0.0336) | (0.0501) |
| Mean (2000) | 167.89 | 64.14 | 11.56 | 2.48 |
| # of Obs. | 5,220 | 5,220 | 5,220 | 5,170 |
| **Panel B. Periphery** | | | | |
| NTR gap × Post | -0.0234 | -0.0368** | -0.0783*** | -0.160*** |
| | (0.0158) | (0.0163) | (0.0248) | (0.0559) |
| Mean (2000) | 16.99 | 5.54 | 0.57 | 0.05 |
| # of Obs. | 15,150 | 15,150 | 15,150 | 12,050 |
| **Panel C. Difference** | | | | |
| Core − Periphery | 0.145*** | 0.127*** | 0.142*** | 0.379*** |
| | (0.0392) | (0.0362) | (0.0418) | (0.0751) |
| County FE | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | ✓ | ✓ | ✓ | ✓ |
| Trends | ✓ | ✓ | ✓ | ✓ |

*Notes:* This table presents heterogeneous effects of TPU reduction on manufacturing outcomes. Counties are ranked by their pre-treatment manufacturing employment share and divided into quartiles; regressions are estimated separately for core counties (top quartile) and periphery counties (remaining quartiles). All specifications are estimated using PPML, with standard errors clustered at the county level. Standard errors of the differences are computed as the square root of the sum of the squared standard errors, since the core and periphery samples are disjoint. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

distribution, and most counties grew regardless of their initial status.[8] At the same time, the ordering within each group reproduces the estimated gradients without controls: employment

[8]More than 95 percent of counties recorded higher manufacturing sales in 2007 than in 1998, and more than 90 percent recorded a larger number of manufacturing firms. The share is lower for employment (69.8 percent), which fell nationwide during the state-sector restructuring of the late 1990s before recovering, and for exports (61.7 percent), where many periphery counties began and ended the period with no recorded exports.

in the more-exposed half of the core rose by 78 percent between 1998 and 2007, compared with 6 percent in the less-exposed half; in the periphery, employment rose by 41 percent in the more-exposed half, compared with 50 percent in the less-exposed half. Therefore, the TPU reduction was far from a uniform stimulus: It reallocated manufacturing activity across space, deepening the pre-existing concentration of manufacturing activities.

To elucidate the mechanisms underlying this spatial reallocation, I next decompose the estimated effects into their extensive and intensive margins. Specifically, does the divergence between core and periphery counties stem from the net entry and exit of firms, or does it reflect the expansion and contraction of incumbents? To answer this, I examine the extensive margin using the total number of firms, further distinguishing between exporting and non-exporting firms, and the intensive margin using average employment, sales, and exports per firm.

The decomposition is also theoretically motivated. In the canonical Melitz framework with Pareto-distributed firm productivity, trade-induced reallocation operates exclusively along the extensive margin: a reduction in variable trade costs triggers firm entry and exit, while average exports per firm are invariant because selection and within-firm expansion offset one another (Melitz, 2003; Chaney, 2008). Examining the relative contribution of these margins therefore provides a natural way to assess whether the empirical adjustment is consistent with this mechanism. In particular, a dominant extensive-margin response accompanied by a stable intensive margin would accord with the predictions of the heterogeneous-firm model with Pareto productivity. Applied to the spatial dimension, this mechanism implies that the divergence between core and periphery counties should arise primarily through differential firm entry and exit rather than changes in average firm size.

Table 3: Extensive and Intensive Margins

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Extensive Margin | | | Intensive Margin | | |
| | # of Firms total | # of Firms exporting | # of Firms non-exporting | Employment per firm | Sales per firm | Exports per firm |
| **Quartiles of Manufacturing Employment Share** | | | | | | |
| **Panel A. Core** | | | | | | |
| NTR gap × Post | 0.122*** | 0.122*** | 0.120*** | 0.00442 | -0.0211 | 0.0205 |
| | (0.0359) | (0.0350) | (0.0444) | (0.0172) | (0.0339) | (0.0397) |
| Mean (2000) | 167.89 | 59.02 | 108.87 | 0.490 | 0.062 | 0.009 |
| # of Obs. | 5,220 | 5,170 | 5,220 | 5,199 | 5,199 | 5,157 |
| **Panel B. Periphery** | | | | | | |
| NTR gap × Post | -0.0234 | -0.170*** | -0.0111 | -0.0142 | 0.00491 | -0.0428 |
| | (0.0158) | (0.0503) | (0.0164) | (0.0131) | (0.0228) | (0.0536) |
| Mean (2000) | 16.99 | 2.14 | 14.85 | 0.335 | 0.034 | 0.002 |
| # of Obs. | 15,150 | 12,050 | 15,150 | 14,844 | 14,844 | 11,958 |
| **Panel C. Difference** | | | | | | |
| Core − Periphery | 0.145*** | 0.292*** | 0.131*** | 0.0186 | -0.0261 | 0.0633 |
| | (0.0392) | (0.0613) | (0.0473) | (0.0216) | (0.0408) | (0.0667) |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Trends | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes:* This table presents the effects of TPU reduction on manufacturing outcomes along extensive and intensive margins. Columns (1)–(3) report the total number of firms and the numbers of exporting and non-exporting firms, respectively. Columns (4)–(6) report average employment, sales, and exports per firm. Counties are ranked by their pre-treatment manufacturing employment share and divided into quartiles; regressions are estimated separately for core counties (top quartile) and periphery counties (remaining quartiles). All specifications are estimated using PPML, with standard errors clustered at the county level. Standard errors of the differences are computed as the square root of the sum of the squared standard errors, since the core and periphery samples are disjoint. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3 decomposes the estimated county-level responses into extensive and intensive margins. Consistent with the Melitz-Pareto framework, adjustment occurs almost exclusively along the extensive margin. In contrast, the coefficients on the intensive margins (Columns 4–6) are statistically indistinguishable from zero. The same asymmetry emerges in the comparison

between the two groups: the core–periphery difference is significant for all three extensive-margin outcomes but not for any of the three intensive-margin outcomes. These results indicate that the spatial reallocative effects of TPU reduction are driven primarily by firm entry and exit, with entry concentrated in core counties and exit concentrated in periphery counties.

The pattern of firm churn in the periphery is particularly revealing. While TPU reduction leads to a substantial decline in the number of exporting firms, this contraction is not accompanied by a corresponding increase in the number of non-exporting firms. This pattern suggests that exiting exporters either left the market or shifted toward domestic sales, potentially crowding out incumbent non-exporters and leaving the total number of non-exporting firms largely unchanged. Because the ASIF covers only above-scale firms, however, a decline in the recorded number of firms may reflect market exit, a fall below the reporting threshold, or relocation to another county, which the aggregate data cannot distinguish. As in the core, the intensive-margin coefficients in periphery counties are small and statistically insignificant.

## 5.3 Uneven Structural Change

I next examine how TPU reduction reshaped the structural composition of local economies. To do so, I construct county-level measures of industrial sophistication by combining firm-level data from the ASIF with industry-level metrics of complexity and upstreamness (Simoes and Hidalgo, 2011; Antràs et al., 2012; Chor et al., 2021). Complexity captures the productive capabilities required to produce a good, while upstreamness measures the average distance of an industry from final demand. Both metrics are aggregated to the county level using employment shares as weights. To examine shifts in manufacturing composition, I classify industries into four

Table 4: Uneven Structural Change within Manufacturing

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| | Complexity | Upstreamness | | Share of Employment (within Manufacturing) | | | |
| | | Mean | Range | Food | Light Manu. | Heavy Manu. | Machinery |
| **Quartiles of Manufacturing Employment Share** | | | | | | | |
| **Panel A. Core** | | | | | | | |
| NTR gap × Post | -0.0213 | 0.0232 | 0.0598** | 0.0600 | -0.791 | 1.43** | -0.697 |
| | (0.0134) | (0.0159) | (0.0279) | (0.409) | (0.702) | (0.713) | (0.518) |
| Mean (2000) | -0.0404 | 3.449 | 3.372 | 8.94 | 27.43 | 40.02 | 23.62 |
| # of Obs. | 5,199 | 5,199 | 5,199 | 5,199 | 5,199 | 5,199 | 5,199 |
| **Panel B. Periphery** | | | | | | | |
| NTR gap × Post | -0.0503*** | -0.00136 | -0.0115 | 1.81*** | -2.62*** | 1.84*** | -1.03*** |
| | (0.0086) | (0.0120) | (0.0197) | (0.432) | (0.456) | (0.531) | (0.292) |
| Mean (2000) | -0.272 | 3.398 | 2.310 | 21.52 | 20.15 | 48.45 | 9.88 |
| # of Obs. | 14,840 | 14,840 | 14,840 | 14,841 | 14,841 | 14,841 | 14,841 |
| **Panel C. Difference** | | | | | | | |
| Core − Periphery | 0.0290* | 0.0246 | 0.0713** | -1.75*** | 1.82** | -0.411 | 0.338 |
| | (0.0159) | (0.0199) | (0.0341) | (0.595) | (0.837) | (0.889) | (0.595) |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Trends | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes:* This table reports the effects of TPU reduction on the composition of local manufacturing. Complexity is the county's employment-weighted average of industry-level product complexity (Simoes and Hidalgo, 2011), with industry complexity fixed at its 1998 value. Upstreamness is measured using China's 2007 input-output tables (Chor et al., 2021), following Antràs et al. (2012). Column (2) reports the county's employment-weighted average upstreamness, while Column (3) reports its range across active industries. Industry values are fixed over time, so variation in Columns (1)–(3) reflects changes in the county's industry mix. Counties are divided into quartiles by pre-treatment manufacturing employment share; regressions are estimated separately for core counties (top quartile) and periphery counties (remaining quartiles). All specifications use OLS with county-clustered standard errors. Columns (4)–(7) report employment shares in percentage points; industry categories are defined in Table A3. Standard errors for core–periphery differences are computed as the square root of the sum of squared standard errors, since the samples are disjoint. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

broad categories: food processing, light manufacturing, heavy manufacturing, and machinery. I supplement this analysis with data from the 2000 and 2010 Chinese National Population Census to capture broader shifts in sectoral employment shares beyond manufacturing.

Table 4 presents the results on structural change within the manufacturing sector. Columns (1) to (3) focus on complexity and upstreamness. The results indicate that TPU reduction exac-

erbated the pre-existing industrial divide. Periphery counties, which were already specialized in less complex industries relative to the core, experience a further shift toward lower-complexity activities. By contrast, core counties exhibit a distinct trajectory: while their average complexity remains stable, TPU reduction significantly expands their upstreamness range. This expansion suggests that core counties diversified their industrial base, developing capabilities to produce a wider spectrum of goods spanning diverse stages of the value chain.

Columns (4) to (7) further unpack these dynamics by examining changes in industry composition within manufacturing. In periphery counties, greater exposure to TPU reduction is associated with a significant reallocation of employment shares from machinery and light manufacturing toward food processing, a structural shift that directly contributes to the observed decline in average complexity. Notably, summary statistics reveal that these regions were historically specialized in food processing while having a minimal presence in machinery prior to PNTR. Consequently, TPU reduction serves to reinforce rather than reverse existing spatial specialization patterns. Instead of fostering convergence, the shock appears to have catalyzed a regression toward static comparative advantage in the periphery, facilitating industrial upgrading in the core while locking lagging regions into lower-complexity activities.

I further investigate whether the spatial reallocative effects of TPU reduction extend beyond the manufacturing sector to influence the broader pattern of structural change. Table 5 reports regression results using county-level sectoral employment data from the Population Census.[9] Columns (1) and (2) focus on population and total employment levels. The results consistently point to a core-periphery divide: TPU reduction generates significant gains in both population

[9]Because the Population Census is conducted decennially, this analysis relies on a two-period panel (2000 and 2010), which accounts for the substantially smaller number of observations relative to the ASIF-based regressions.

Table 5: Uneven Structural Change across Sectors

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Population | Employment | Share of Employment | | | |
| | | | Agriculture | Manufacturing | Other Industry | Services |
| **Quartiles of Manufacturing Employment Share** | | | | | | |
| **Panel A. Core** | | | | | | |
| NTR gap × Post | 0.0309** | 0.0554*** | -0.155 | 0.755* | -0.153 | -0.431 |
| | (0.0126) | (0.0143) | (0.590) | (0.433) | (0.234) | (0.287) |
| Mean (2000) | 935.86 | 48.54 | 45.02 | 20.40 | 7.45 | 27.13 |
| # of Obs. | 1,044 | 1,044 | 1,042 | 1,044 | 1,042 | 1,042 |
| **Panel B. Periphery** | | | | | | |
| NTR gap × Post | -0.00231 | 0.00136 | -0.481** | 0.174 | 0.133 | 0.188 |
| | (0.00314) | (0.00420) | (0.218) | (0.106) | (0.0927) | (0.125) |
| Mean (2000) | 477.40 | 26.28 | 79.20 | 4.22 | 3.14 | 13.45 |
| # of Obs. | 3,020 | 3,020 | 3,016 | 3,016 | 3,016 | 3,016 |
| **Panel C. Difference** | | | | | | |
| Core − Periphery | 0.0332** | 0.0539*** | 0.326 | 0.580 | -0.286 | -0.619** |
| | (0.0130) | (0.0149) | (0.629) | (0.446) | (0.252) | (0.313) |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Trends | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes:* This table presents the effects of TPU reduction on the pattern of structural change across sectors. Counties are ranked by their pre-treatment manufacturing employment share and divided into quartiles; regressions are estimated separately for core counties (top quartile) and periphery counties (remaining quartiles). Columns (1) and (2) report population and employment (thousands) and are estimated using PPML. Columns (3) to (6) report employment shares (percentage points) and are estimated using OLS. Population is measured from the complete census, whereas employment is based on the census long form, which covers a 10 percent sample of households. Other Industry comprises non-manufacturing activities in the secondary sector, including mining, utilities, and construction. Standard errors are clustered at the county level. Standard errors of the differences are computed as the square root of the sum of the squared standard errors, since the core and periphery samples are disjoint. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

and employment in core counties, while periphery counties show no response on either margin. Columns (3) to (6) examine the effects on inter-sectoral structural change. Consistent with Erten and Leight (2021), lower TPU shifts labor out of agriculture, with substantially larger and more precisely estimated effects in periphery counties. However, the destination of this reallocation differs sharply across the two groups. In core counties, manufacturing expands significantly and

serves as the dominant destination for displaced agricultural labor. In periphery counties, by contrast, none of the three non-agricultural sectors shows a significant increase. Thus, greater exposure produces local industrialization in the core but agricultural exit without corresponding local employment growth in the periphery.[10] Moreover, the positive manufacturing coefficient is substantially smaller in magnitude than the negative agricultural coefficient, indicating that manufacturing growth offsets only a limited share of the agricultural decline.

This asymmetry is consistent with the role of internal migration. Faced with limited local industrial capacity, workers leaving agriculture in periphery counties are likely migrating to core counties where manufacturing sectors are expanding. Consequently, structural change in periphery areas occurs without corresponding local industrial development. Therefore, while TPU reduction facilitates manufacturing growth in the core, it does not generate comparable industrial growth in the periphery and may even constrain local development by reinforcing out-migration. These findings motivate the model developed in Section 7, which incorporates internal migration and regional productivity differences as the two core mechanisms rationalizing the spatially heterogeneous effects of trade integration.

## 5.4 Pre-trends

The causal interpretation of the heterogeneous effects documented above rests on the parallel trends assumption. In the standard DID framework, this requires that counties with different exposure levels would have followed similar trajectories in manufacturing outcomes absent the TPU reduction. Given the sample splitting strategy employed in the main analysis, the validity

[10] Using rural household panel data, Ma et al. (2025) likewise find that greater exposure increases the number and share of households exiting agriculture entirely. Their measures, however, do not identify where or in which sector household members subsequently work.

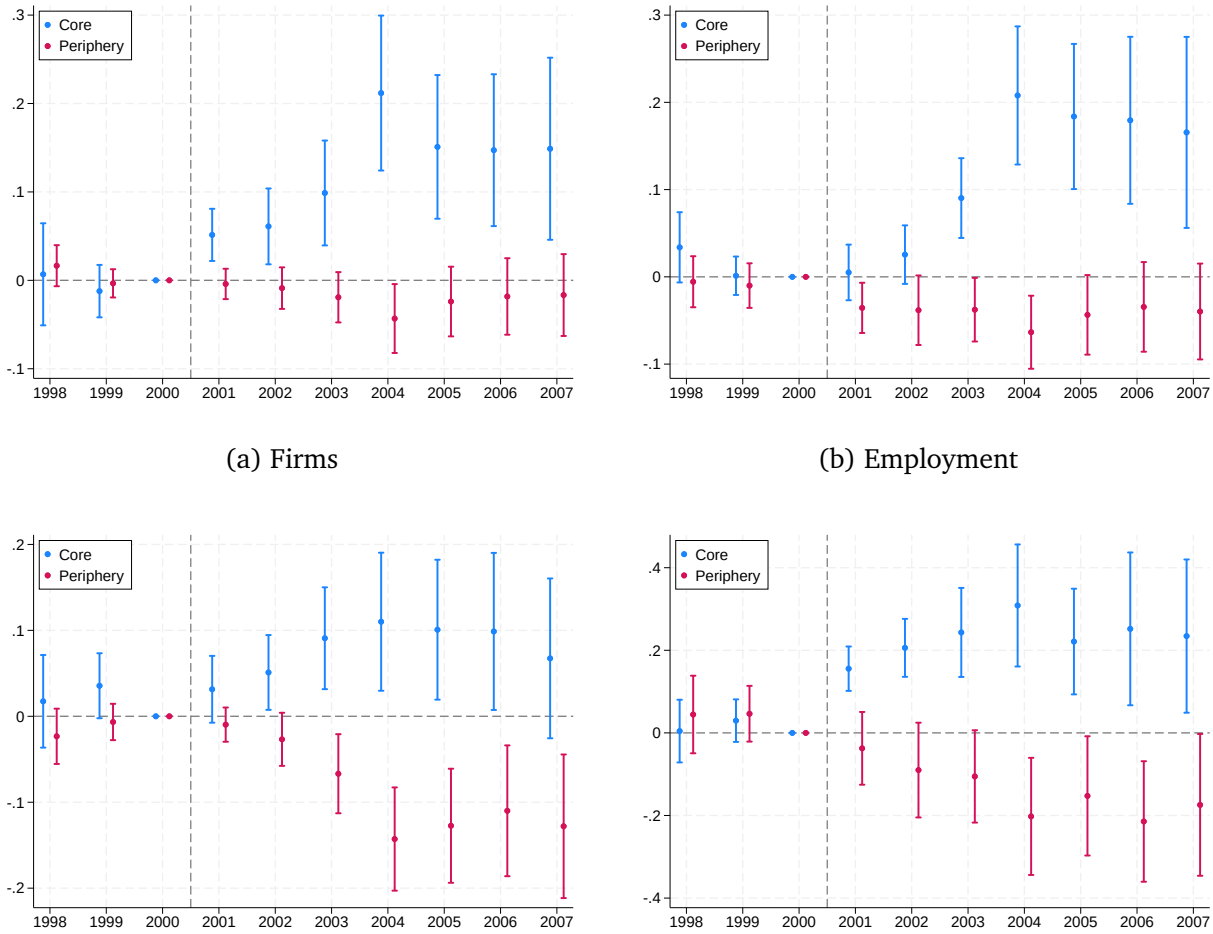
of the estimates depends specifically on whether this assumption holds *within* the core and periphery subsamples, respectively. That is, identification is derived from comparing high- and low-exposure counties within each geographic group. Consequently, examining pre-treatment dynamics serves a dual purpose: it validates the internal validity of the estimates for each group, and by comparing the estimated trajectories side-by-side, it allows us to assess whether the spatial divergence is indeed a distinct post-shock phenomenon.

Although the parallel trends assumption cannot be directly tested, pre-treatment trends provide evidence on its plausibility. I estimate the following event study specification separately for the core and periphery subsamples to test for pre-existing differential trends and trace the dynamic evolution of treatment effects:

$$Y_{ct} = \exp\left[\sum_{\tau \neq 2000} \beta_\tau \cdot (\text{NTR Gap}_c \times \mathbb{1}_{\{t=\tau\}}) + \delta_c + \lambda_t + \mathbf{X}'_{ct}\boldsymbol{\gamma} + \mathbf{Z}'_c t\right] + \varepsilon_{ct}, \tag{5}$$

where $\text{NTR Gap}_c$ denotes the local exposure to TPU reduction, and $\mathbb{1}_{\{t=\tau\}}$ is a set of time dummy variables for each year $\tau \in \{1998, \ldots, 2007\}$. The year 2000 serves as the base period, so its interaction term is omitted. The coefficients of interest, $\beta_\tau$, capture the dynamic effects of TPU exposure on local manufacturing outcomes relative to the base year within each subsample. All other terms are defined as in equation (4).

Figure 4 plots the group-specific event-study estimates for four outcomes, while Table A8 reports formal tests of the pre-treatment coefficients and of the differences between the core and periphery paths. In the pre-treatment period, the estimates for 1998 and 1999 are generally small, and joint tests fail to reject that the two coefficients are jointly zero at the five percent level for all eight group–outcome pairs. The weakest cases are employment and sales in the




(a) Firms

(b) Employment

(c) Sales

(d) Exports

Figure 4: Event Study Plots

*Notes:* This figure plots the event study coefficient estimates with 95% confidence intervals. Counties are ranked by their pre-treatment manufacturing employment share and divided into quartiles; regressions are estimated separately for core counties (top quartile) and periphery counties (remaining quartiles). All specifications are estimated using PPML, with standard errors clustered at the county level. The year 2000 serves as the base period.

core, with $p$-values of 0.079 and 0.071, respectively, although neither annual coefficient is individually significant. More directly relevant to the core–periphery comparison, the pre-treatment differences are jointly insignificant for every outcome, with $p$-values ranging from 0.150 to 0.875. Of the eight annual pre-treatment differences, only the 1999 estimate for sales is individually significant, and only at the ten percent level. After the policy change, the two paths separate: more-exposed core counties move above their less-exposed counterparts, while more-exposed periphery counties fall below theirs. For all four outcomes, joint tests reject the null that the post-treatment core–periphery coefficient differences are zero ($p < 0.001$). The divergence therefore emerges after PNTR rather than reflecting systematic differential trends before the policy change.

## 5.5 Assessing the Shift-Share Design

This section assesses the validity of the shift-share design used to construct county-level NTR gaps. As discussed in Section 4.1, identification in this setting primarily relies on the exogeneity of the shift component (shocks), since the industry-level NTR gaps are determined by historical tariffs set long before China's WTO accession. I follow the framework and empirical diagnostics developed by Borusyak et al. (2022) for shift-share designs with exogenous shocks.

**(In)complete Shares.** The county-level NTR gap is constructed using employment shares *within* manufacturing, so the shares sum to one in every county and the design falls into the "complete shares" case of Borusyak et al. (2022). This raises the concern that non-manufacturing industries, which employed the majority of workers before China's WTO accession, are excluded from the exposure measure. Table A12 compares the identifying variation under the two

definitions. Rescaling the manufacturing shares by the county manufacturing employment share and assigning the residual to a non-manufacturing sector with a zero NTR gap reduces the importance-weighted interquartile range from 28.4 percentage points to zero. At the same time, it reduces the effective number of shocks, computed as the inverse of the Herfindahl–Hirschman Index, from 42.8 to 1.2. The unshocked non-manufacturing sector alone accounts for 80 percent of the importance weight in the core and 96 percent in the periphery. I therefore retain the within-manufacturing definition, following Kovak (2013), who argues that non-traded activities should be excluded from the weighted average rather than assigned a zero shock.

**Correlation across Shocks.** I next examine the extent of correlation across industry-level shocks and whether such correlation affects inference. Table A13 decomposes the variance of four-digit industry NTR gaps into components at the two- and three-digit industry levels and a four-digit-specific residual. The two- and three-digit components account for 33.4 and 23.4 percent of the total variation, respectively, indicating substantial common variation in NTR gaps among industries belonging to the same broader industry groups. This pattern is consistent with the historical structure of the Smoot–Hawley tariff schedule, under which related products within broader industries were often subject to similar levels of protection. To address this concern, I report exposure-robust standard errors in Table A14, following Adao et al. (2019). I first treat the 425 four-digit industries as independent shocks and then allow shocks to be arbitrarily correlated within the 29 two-digit industries. The substantive conclusions are robust to both inference procedures.

**Placebo Tests.** Finally, I conduct a placebo test to assess whether the estimated effects could be driven by spurious correlations. First, I randomly permute county-level NTR gaps, preserving their empirical distribution while holding all other county-level characteristics fixed. Second, I randomly permute four-digit-industry-level NTR gaps and reconstruct county-level exposure using the original employment shares. For each permutation, I re-estimate the baseline specification. Figures A5 and A6 plot the resulting distributions of placebo estimates from the county- and industry-level permutations, respectively. In both exercises, the placebo estimates are centered around zero. More importantly, the estimated core–periphery differences lie toward the tails of their corresponding placebo distributions across all four outcomes, indicating that the documented spatial divergence is unlikely to arise from random reassignment of either county-level exposure or the underlying industry-level shocks.

## 5.6 Robustness Checks

This section presents empirical evidence supporting the robustness of the main results against alternative specifications, sample restrictions, estimation methods, and variable definitions.

**Alternative Fixed Effects.** First, I assess whether the results are driven by omitted time-varying local characteristics. I replicate the baseline regressions from Table 2 using a more stringent specification that replaces regional time trends with province-by-year fixed effects. Additionally, I include interaction terms between year fixed effects and key initial county characteristics, including administrative status, distance to coast and rivers, and the pre-treatment SOE share. As reported in Table A9, absorbing these time-varying local shocks does not alter the main findings regarding the spatial reallocative effects of TPU reduction. The coefficient estimates

remain consistent in sign and magnitude, and most retain statistical significance, with the exception of sales in core counties.

**Sensitivity to Outliers.** Second, I test whether the main results are dominated by counties at the extremes of the exposure distribution. I rank counties by their NTR gaps and re-estimate the model after trimming the top and bottom 1, 5, and 10 percent of the sample within each group. Table A10 presents the results. Although the reduction in sample size naturally widens the confidence intervals, the main findings remain qualitatively unchanged. In several cases, the estimated coefficients are even larger in magnitude than the baseline. These patterns confirm that the spatial reallocation effects are a general phenomenon across the sample distribution rather than an artifact of a few outlier counties.

**Alternative Estimation Methods.** Third, I test the sensitivity of the results to functional form assumptions by comparing the baseline PPML estimator with OLS and log-linear specifications. For each outcome variable $Y$, I estimate three alternative models: $\log(Y)$ using OLS, $\log(Y)$ using WLS, and $\log(c+Y)$ using WLS, where $c$ is set to the minimum observed positive value of $Y$. Weights for WLS are based on pre-treatment manufacturing employment (1998–2000) to account for heteroskedasticity related to economic size. As shown in Table A11, the coefficients largely preserve the signs of the baseline estimates, with most remaining statistically significant. However, for exports, the exclusion of zero flows in the log specification and the sensitivity of $\log(c+Y)$ to the choice of $c$ lead to implausibly large estimates for periphery counties. This instability highlights the advantage of the PPML estimator in handling zero trade flows and heteroskedasticity, further justifying its use as the baseline method.

**Alternative Quantile Definitions.** Finally, I verify that the observed results do not depend on the quartile cutoff used to define the core. I re-partition the sample into 6, 8, and 10 quantiles and, in each case, compare the top quantile with all remaining counties. Figure A4 reports the estimates. As the definition of the core becomes more exclusive, the exposure gradient within the top quantile remains large, positive and significant for all outcomes. Meanwhile, the gradient among the remaining counties remains negative or indistinguishable from zero. The positive gradient therefore remains concentrated among the most industrialized counties even as the core is defined narrowly, while no positive gradient appears in the remainder of the sample. Therefore, the core–periphery divergence reflects the amplification of pre-existing differences in industrial capacity rather than sensitivity to a particular cutoff.

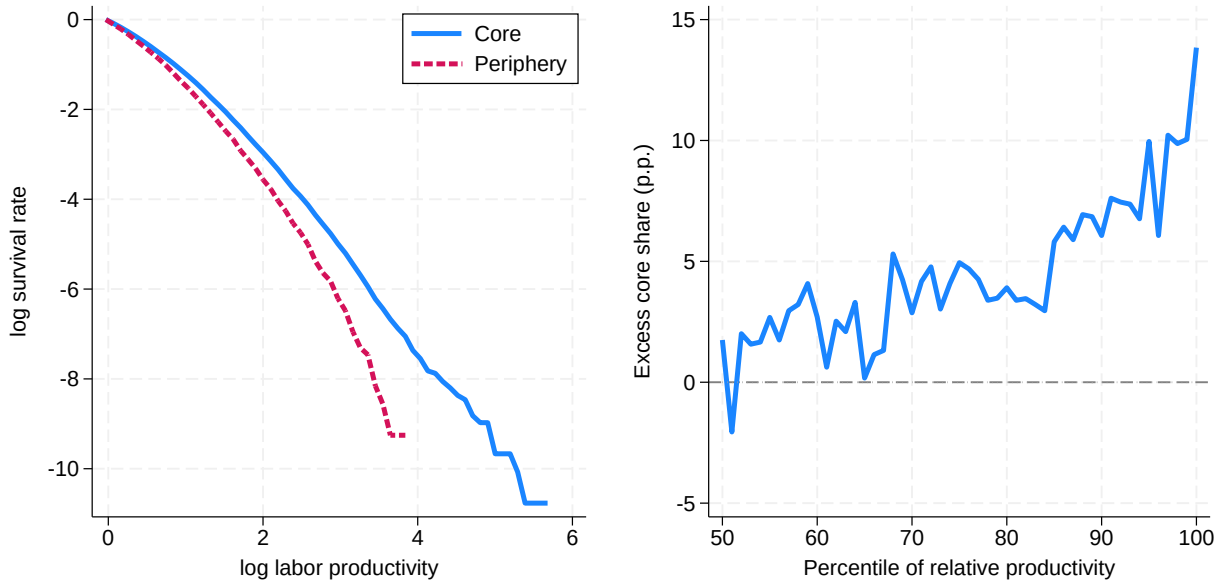
# 6 Mechanisms

The spatial reallocation documented above suggests two underlying mechanisms: regional productivity differences and internal migration. In this section, I examine each mechanism in turn, providing empirical evidence on productivity distributions and migration patterns.

## 6.1 Regional Productivity Differences

A major source of gains from trade arises from improving allocative efficiency within industries, as market share shifts toward more productive firms (Melitz, 2003). When high-productivity firms are spatially concentrated, even a uniform reduction in TPU can generate uneven local responses. Moreover, under the standard assumption of Pareto-distributed firm productivity, trade-induced adjustment is predicted to occur primarily along the extensive margin, a pattern

consistent with the results in Table 3. However, does the observed spatial heterogeneity truly stem from underlying differences in firm-level productivity distributions?




(a) Pareto Tails

(b) Excess Core Share

Figure 5: Productivity Distributions and Core–Periphery Asymmetry

*Notes:* This figure compares firm productivity distributions between core and periphery counties in the year 2000. Productivity is measured by log labor productivity, defined as $\log(Y/L)$, and is normalized by the industry-level geometric mean. Panel (a) plots the log survival function and highlights the approximately linear relationship in the upper tail. Panel (b) plots the excess core share by productivity percentile, showing increasing over-representation of core firms toward the upper tail relative to their baseline share (77.25%).

Figure 5 documents systematic differences in productivity distributions between core and periphery counties using the simplest productivity measure: log labor productivity $\log(Y/L)$. Both panels focus on the upper half of the productivity distribution, where differences in tail behavior are most pronounced. Panel (a) plots the log survival function against log labor productivity. While the distribution is curved at lower productivity levels, an approximately linear relationship emerges over a substantial portion of the upper tail, consistent with a Pareto distribution. Within this linear region, the core exhibits a flatter slope, implying a smaller Pareto

Table 6: Pareto Shape Estimates by Tail Cutoff, 1998–2000

| | (1) All | (2) Top 50% | (3) Top 25% | (4) Top 10% | (5) Top 5% | (6) Top 3% | (7) Top 1% |
|---|---|---|---|---|---|---|---|
| **Panel A. Core** | | | | | | | |
| $\alpha_C$ | 2.717 | 3.223 | 3.508 | 3.730 | 3.837 | 3.905 | 4.099 |
| | (0.009) | (0.017) | (0.027) | (0.044) | (0.062) | (0.079) | (0.123) |
| # of Obs. | 262,909 | 131,455 | 65,727 | 26,291 | 13,145 | 7,887 | 2,629 |
| **Panel B. Periphery** | | | | | | | |
| $\alpha_P$ | 3.182 | 3.871 | 4.307 | 4.722 | 4.954 | 5.138 | 5.740 |
| | (0.018) | (0.036) | (0.060) | (0.106) | (0.149) | (0.192) | (0.337) |
| # of Obs. | 77,029 | 38,515 | 19,257 | 7,703 | 3,851 | 2,311 | 770 |
| **Panel C. Difference** | | | | | | | |
| $\alpha_C - \alpha_P$ | -0.465*** | -0.648*** | -0.799*** | -0.992*** | -1.117*** | -1.233*** | -1.641*** |
| | (0.020) | (0.040) | (0.066) | (0.115) | (0.162) | (0.207) | (0.359) |

*Notes:* This table reports Pareto shape parameters from QQ estimation on firm sales, estimated separately for core and periphery counties over the pre-accession window 1998–2000. "Top X%" retains firms above the Xth percentile of the sales distribution within the corresponding group. The shape parameter is obtained by mapping the estimated QQ slope under a trade elasticity of $\sigma = 4$. Standard errors are obtained from 500 firm-level block bootstrap replications, resampling firms with replacement and retaining all yearly observations for each sampled firm. Standard errors of the differences are computed as the square root of the sum of the squared bootstrap standard errors, since the core and periphery samples are disjoint. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

shape parameter and a thicker upper tail of high-productivity firms relative to the periphery. Panel (b) complements this evidence by examining the composition of firms along the upper end of the productivity distribution. Ranking firms by relative productivity in the full sample, the figure shows that the excess share of core firms increases monotonically across higher productivity percentiles. Together, these patterns confirm that the core's advantage stems not merely from having more firms, but from its specific capacity to generate and retain the most productive firms in the economy.

To discipline the theoretical shape parameters and formally quantify differences in tail thickness, I estimate the Pareto shape parameters for firm productivity in core and periphery counties using the QQ estimation approach of Head et al. (2014) and Bakker et al. (2024). The estimation relies on the linear relationship between empirical quantiles of log sales and

theoretical Pareto quantiles, where the slope corresponds to the inverse of the Pareto shape parameter, scaled by the trade elasticity. Table 6 reports the estimates for the pre-accession window 1998–2000, under the assumption that the trade elasticity is $\sigma = 4$. Across specifications, the implied Pareto shape parameter is consistently smaller for the core than for the periphery, indicating a thicker upper tail of highly productive firms in core regions. This finding is consistent with the patterns documented in the graphical evidence.[11]

A natural alternative is that population agglomeration, rather than industrial capability, separates the two groups. As Table A17 shows, density does complement growth in the core, as classic agglomeration theories would predict (Marshall, 1920). In the periphery, however, higher density is associated with weaker sales and exports. When density and the initial manufacturing employment share are included jointly in the full-sample regression, the density interaction no longer exhibits the positive effect, while the manufacturing-share interaction remains positive and significant across all outcomes. The core–periphery divide therefore appears to reflect the strength of the local manufacturing base rather than population agglomeration *per se*, consistent with the tail-thickness evidence documented above. Without such a manufacturing base, density alone need not generate local expansion: a larger population may instead provide labor that is drawn to the core through internal migration, the mechanism I turn to next.

[11] Tables A15 and A16 repeat the estimation separately for 1998–2000, 2001–2003, and 2005–2007, using all firms and entrants, respectively. The core exhibits a significantly thicker tail at every cutoff in each period, although the gap narrows slightly over time. The entrant results link this tail-thickness gap to the extensive-margin adjustment documented in Table 3: firms entering core counties are more concentrated in the upper tail of the productivity distribution than those entering periphery counties.

## 6.2 Internal Migration

Previous studies document that Chinese prefectures more exposed to TPU reduction experience higher in-migration rates (Facchini et al., 2019). With county-level data, I ask three questions in turn: whether this pattern persists at the county level, which margin of migration responds, and whether manufacturing outcomes in periphery counties deteriorate with exposure transmitted through historical migration networks. Throughout the analysis, I distinguish between two treatment variables. A county's own NTR gap captures the export-demand shock to its manufacturers, whereas its migration-network exposure captures the pull generated by pre-existing connections to destinations receiving larger shocks. As the estimates below show, these two forces operate through different margins.

I characterize pre-existing migration networks using data from the 2000 Population Census, which records respondents' residence in 1995. Based on these historical migration flows, I compute a measure of *indirect* exposure to TPU reduction, defined as the migration-weighted average of NTR gap reductions in destination counties.[12]

Table 7 presents the effects of TPU reduction on county-level migration outcomes. In core counties, higher NTR gaps significantly increase both the net migration balance and the share of residents registered outside the county. The latter increase is driven predominantly by cross-province, non-*hukou* migrants, consistent with Facchini et al. (2019). However, their findings capture only one side of the spatial adjustment. In periphery counties, no such pattern emerges: a county's own exposure to TPU reduction is unrelated to any of these migration outcomes. This contrast is statistically significant for the net migration balance and cross-province registrations.

[12]Migration weights are defined as the share of migrants from each origin prefecture moving to each destination prefecture between 1995 and 2000. Since the census identifies migrants' previous residence only at the prefecture level, I measure the destination shock using the NTR gap of the merged urban districts within each prefecture.

Table 7: Exposure and County-Level Migration, 2000–2010

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| | Net migration balance | | Residents registered outside the county | | | | | |
| | | | Total | | Cross-province | | Within-province | |
| **Quartiles of Manufacturing Employment Share** | | | | | | | | |
| **Panel A. Core** | | | | | | | | |
| NTR gap × Post | 3.899** | 3.763** | 5.343*** | 5.156*** | 4.012*** | 3.924*** | 1.331* | 1.232 |
| | (1.725) | (1.704) | (1.820) | (1.775) | (1.361) | (1.348) | (0.792) | (0.777) |
| NTR gap (Mig. Network) × Post | | -5.274 | | -7.270* | | -3.406 | | -3.864** |
| | | (4.024) | | (4.376) | | (3.416) | | (1.740) |
| # of Obs. | 1,042 | 1,042 | 1,042 | 1,042 | 1,042 | 1,042 | 1,042 | 1,042 |
| **Panel B. Periphery** | | | | | | | | |
| NTR gap × Post | -0.119 | -0.041 | 0.030 | 0.033 | -0.014 | -0.014 | 0.044 | 0.047 |
| | (0.255) | (0.254) | (0.081) | (0.081) | (0.034) | (0.034) | (0.061) | (0.060) |
| NTR gap (Mig. Network) × Post | | -2.409*** | | -0.117 | | 0.003 | | -0.120 |
| | | (0.528) | | (0.215) | | (0.088) | | (0.162) |
| # of Obs. | 3,018 | 3,018 | 3,018 | 3,018 | 3,018 | 3,018 | 3,018 | 3,018 |
| **Panel C. Difference** | | | | | | | | |
| Core − Periphery (Own) | 4.018** | 3.804** | 5.313*** | 5.123*** | 4.026*** | 3.938*** | 1.288 | 1.184 |
| | (1.743) | (1.723) | (1.822) | (1.777) | (1.361) | (1.349) | (0.794) | (0.779) |
| Core − Periphery (Network) | | -2.865 | | -7.153 | | -3.409 | | -3.744** |
| | | (4.058) | | (4.382) | | (3.417) | | (1.748) |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Trends | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes:* This table presents the effects of TPU reduction on county-level migration outcomes using the county tabulations of the 2000 and 2010 Population Censuses. Columns (1) and (2) report the net migration balance, the resident population less the county's registered (*hukou*) population; a positive value means residents in excess of registrations. Columns (3) to (8) count residents whose *hukou* is registered outside the county, in total and separately for registrations in another province and in another county of the same province. Even-numbered columns additionally control for indirect exposure through pre-existing migration networks, measured as the migration-weighted average of destination NTR gaps constructed from the 1995–2000 migration flows. All outcomes are expressed per hundred of the county's 2000 registered population, and regressions are weighted by that population. Counties are ranked by their pre-treatment manufacturing employment share and divided into quartiles; regressions are estimated separately for core counties (top quartile) and periphery counties (remaining quartiles). All specifications use the same fixed effects, tariff controls, and trends as Table 2, with standard errors clustered at the county level. Standard errors of the differences are computed as the square root of the sum of the squared standard errors, since the core and periphery samples are disjoint. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

On the other hand, network exposure is associated with a deterioration in the net migration balance, consistent with an external pull toward connected destinations receiving larger trade shocks. The corresponding coefficient is also negative, though imprecisely estimated, in core counties. Taken together, these results reveal two distinct sides of the migration response: direct exposure attracts migrants to manufacturing-intensive counties, while historical migration links transmit that pull to origin counties that need not receive a large trade shock themselves.

Table 8: Migration-Network Exposure and Local Manufacturing Outcomes

| | (1) Firms | (2) Employment | (3) Sales | (4) Exports |
|---|---|---|---|---|
| **Quartiles of Manufacturing Employment Share** | | | | |
| **Panel A. Core** | | | | |
| NTR gap × Post | 0.130*** | 0.111*** | 0.0743** | 0.227*** |
| | (0.0373) | (0.0342) | (0.0353) | (0.0562) |
| NTR gap (Mig. Network) × Post | -0.0337 | -0.102*** | -0.0505 | -0.0282 |
| | (0.0352) | (0.0313) | (0.0322) | (0.0634) |
| # of Obs. | 5,220 | 5,220 | 5,220 | 5,170 |
| **Panel B. Periphery** | | | | |
| NTR gap × Post | -0.0199 | -0.0309* | -0.0705*** | -0.168*** |
| | (0.0161) | (0.0166) | (0.0254) | (0.0586) |
| NTR gap (Mig. Network) × Post | -0.0299* | -0.0517*** | -0.0525* | 0.0502 |
| | (0.0156) | (0.0168) | (0.0270) | (0.0703) |
| # of Obs. | 15,150 | 15,150 | 15,150 | 12,050 |
| **Panel C. Difference** | | | | |
| Core − Periphery (Own) | 0.150*** | 0.142*** | 0.145*** | 0.395*** |
| | (0.0406) | (0.0380) | (0.0435) | (0.0812) |
| Core − Periphery (Network) | -0.00380 | -0.0500 | 0.00200 | -0.0784 |
| | (0.0385) | (0.0355) | (0.0420) | (0.0947) |
| County FE | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | ✓ | ✓ | ✓ | ✓ |
| Trends | ✓ | ✓ | ✓ | ✓ |

*Notes:* This table presents the effects of TPU reduction after augmenting the baseline specification with indirect exposure through pre-existing migration networks. Migration-network exposure is measured as the migration-weighted average of destination NTR gaps constructed from the 1995–2000 migration flows. Counties are ranked by their pre-treatment manufacturing employment share and divided into quartiles; regressions are estimated separately for core counties (top quartile) and periphery counties (remaining quartiles). All specifications use the same fixed effects, tariff controls, and trends as Table 2, with standard errors clustered at the county level. Standard errors of the differences are computed as the square root of the sum of the squared standard errors, since the core and periphery samples are disjoint. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Migration-network exposure also predicts local manufacturing outcomes. Table 8 augments the baseline specification with indirect exposure through pre-existing migration networks. The core–periphery divergence in response to a county's own exposure remains virtually unchanged. At the same time, greater network exposure significantly reduces manufacturing employment in both groups, consistent with labor being drawn toward historically connected destinations receiving larger trade shocks. In periphery counties, the coefficients on the number of firms and total sales are also negative, although estimated less precisely. Importantly, the network effects do not differ significantly between core and periphery counties. What distinguishes the two groups is therefore their response to their own trade opportunities: core counties expand strongly despite the external pull, whereas periphery counties do not.

Taken together, the migration results point to a spatial adjustment mechanism that differs from a simple push response to local trade shocks. A periphery county's own exposure does not induce greater out-migration. Instead, migration responds to opportunities elsewhere, transmitted through pre-existing network connections to expanding destinations. This external pull is associated with both a deterioration in the local net migration balance and a relative decline in manufacturing employment. Unlike core counties, periphery counties do not generate sufficiently strong manufacturing growth in response to their own trade opportunities to counteract this external pull. Migration therefore reinforces the initial divergence in local adjustment, reallocating labor away from places that fail to expand and toward those better positioned to benefit from trade integration.

# 7 Theoretical Framework

In this section, I develop a spatial general equilibrium model to rationalize the divergent patterns in the empirical analysis: the same reduction in TPU spurred manufacturing expansion in core counties while leading to a relative decline in the periphery. I first clarify the theoretical link between TPU and trade barriers, and then introduce a heterogeneous-firm model with internal labor mobility and region-specific productivity distributions, following Bakker et al. (2024). The framework illustrates how differences in the underlying productivity distribution can generate sharply divergent regional responses to a common reduction in trade barriers.

## 7.1 The Effects of TPU on Real Economic Outcomes

Uncertainty about the future trade environment affects firms' export decisions when market access involves sunk costs and irreversible investments. In such environments, uncertainty generates an option value of waiting: firms postpone export entry and capacity expansion until policy conditions become clearer (Bloom et al., 2007). Elevated TPU therefore depresses export participation and dampens export intensity among active firms.

In the context of U.S.–China trade, Handley and Limão (2017) formalize this mechanism in a dynamic heterogeneous-firm model in which firms incur sunk costs to enter export markets while facing the possibility that current tariff rates may revert to higher levels. The risk of a future tariff increase lowers the expected continuation value of exporting and raises the productivity threshold required for profitable entry. When uncertainty is resolved and the threat of tariff reversal is eliminated, export entry expands and existing exporters scale up.

While the underlying mechanism operates through expectations about future policy states, its long-run real effects can be represented in reduced form as a change in effective trade barriers. In particular, the resolution of TPU reduces the expected trade wedge between Home and Foreign markets. In the model below, I capture this effect as a deterministic decline in the iceberg trade cost of exporting. This reduced-form representation preserves the key comparative-static implications of the uncertainty models—namely, lower export cutoffs and greater export participation—while abstracting from transitional dynamics. It also allows the analysis to focus on how a common reduction in international trade costs propagates across regions with different initial industrial structures and productivity distributions.

## 7.2 Model Setup

This subsection develops a stylized spatial general equilibrium model with agriculture and manufacturing, in which regions differ in the tails of their firm productivity distributions and workers can migrate domestically. I show that trade liberalization generates asymmetric spatial effects, reallocating manufacturing activity toward regions with thicker productivity tails while inducing structural change unevenly across space.

### 7.2.1 Environment

The world consists of two countries, Home ($H$) and Foreign ($F$). Home contains two regions, Core ($C$) and Periphery ($P$), indexed by $i \in \{C, P\}$. Total Home population is fixed at $\bar{L}$, and workers are freely mobile across regions within Home but cannot migrate internationally. Home produces goods in two sectors: agriculture ($A$) and manufacturing ($M$). Agricultural goods are homogeneous and freely traded both domestically and internationally. Manufacturing goods

are a CES aggregate of differentiated varieties produced under monopolistic competition, and are freely traded domestically but subject to iceberg trade costs $\tau > 1$ in international trade. Foreign is treated as exogenous and provides an outside market for Home manufacturing firms.

### 7.2.2 Preferences and Migration

Workers have non-homothetic Stone–Geary preferences and idiosyncratic locational tastes. Specifically, a worker residing in region $i$ attains utility

$$U_i(\omega) = \beta \log(A_i - \bar{A}) + (1-\beta)\log C_i + \log \epsilon_i(\omega), \tag{6}$$

where $\bar{A} > 0$ is a subsistence requirement for agricultural consumption, $C_i$ is a CES aggregate of differentiated manufacturing varieties with elasticity of substitution $\sigma > 1$, and $\epsilon_i(\omega)$ captures idiosyncratic preferences over regions. The manufacturing composite and associated price index are given by

$$C_i = \left( \int_{\omega \in \Omega_i} c_i(\omega)^{\frac{\sigma-1}{\sigma}} \, d\omega \right)^{\frac{\sigma}{\sigma-1}}, \qquad P_i^M = \left( \int_{\omega \in \Omega_i} p_i(\omega)^{1-\sigma} \, d\omega \right)^{\frac{1}{1-\sigma}}. \tag{7}$$

I focus on the limiting case $\beta \to 0$, in which workers consume the subsistence quantity $\bar{A}$ and spend all remaining income on manufacturing. Real disposable income in region $i$ is

$$V_i \equiv \frac{y_i}{P_i^M} = \frac{w_i + r_i \bar{T}_i / L_i - p_A \bar{A}}{P_i^M}, \tag{8}$$

where $w_i$ is the wage, $r_i \bar{T}_i / L_i$ is per-capita land rental income rebated to residents of region $i$, and $p_A \bar{A}$ is the cost of subsistence agricultural consumption.

Idiosyncratic location preferences are i.i.d. Fréchet-distributed with shape parameter $\nu > 1$. Workers choose their location to maximize utility, implying that the equilibrium distribution of population across regions satisfies

$$\ln\left(\frac{L_C}{L_P}\right) = \nu \ln\left(\frac{V_C}{V_P}\right), \tag{9}$$

where $\nu$ governs the elasticity of migration with respect to real income differences.

### 7.2.3 Agriculture

Agricultural goods are produced under perfect competition using land and labor with a Cobb–Douglas technology:

$$Y_i^A = a_A \bar{T}_i^{1-\eta} (L_i^A)^{\eta}, \tag{10}$$

where $\bar{T}_i$ is the fixed land endowment in region $i$, $L_i^A$ is agricultural labor, $a_A$ is agricultural productivity, and $\eta \in (0,1)$ is the labor share. Home can trade agricultural goods freely with Foreign at the world price, which I normalize to $p_A = 1$. This ensures that any gap between domestic production and consumption of agricultural goods is met by imports. Perfect competition implies zero profits, so factor prices satisfy

$$w_i^A = \eta\, a_A \left(\frac{\bar{T}_i}{L_i^A}\right)^{1-\eta}, \qquad r_i = (1-\eta)\, a_A \left(\frac{L_i^A}{\bar{T}_i}\right)^{\eta}, \tag{11}$$

where $r_i$ is the land rental rate. Workers move freely across sectors within a region, so $w_i^A = w_i$. Land rental income $r_i \bar{T}_i$ is rebated lump-sum to all residents of region $i$.

### 7.2.4 Manufacturing

**Production and pricing.** In each region, an unlimited pool of potential entrants pay a sunk entry cost $f^e$ in units of local labor and draw productivity $\varphi$ from a Pareto distribution with support $[1, \infty)$ and region-specific shape parameter $\alpha_i > 1$:

$$G_i(\varphi) = 1 - \varphi^{-\alpha_i}, \qquad \alpha_C < \alpha_P. \tag{12}$$

The assumption $\alpha_C < \alpha_P$ implies that Core has a thicker right tail—i.e., a greater share of high-productivity firms.

After entry, serving the domestic market requires a fixed cost $f^d$ and exporting to Foreign requires an additional fixed cost $f^x > f^d$, both in units of local labor. Under monopolistic competition with CES preferences, a firm with productivity $\varphi$ in region $i$ charges

$$p_{ij}(\varphi) = \frac{\sigma}{\sigma - 1} \frac{\tau_{ij} w_i}{\varphi}, \tag{13}$$

where $\tau_{ij} = \tau > 1$ if $j = F$ and $\tau_{ij} = 1$ otherwise.

**Zero-profit conditions.** Manufacturing expenditure in region $j$ equals aggregate disposable income, $E_j^M = L_j y_j$. For a firm in region $i$ selling to market $j$, CES demand yields revenue

$$r_{ij}(\varphi) = \left( \frac{p_{ij}(\varphi)}{P_j^M} \right)^{1-\sigma} E_j^M, \tag{14}$$

with operating profits equal to $r_{ij}(\varphi)/\sigma$. The domestic and export productivity cutoffs are defined by zero-profit conditions:

$$\frac{1}{\sigma}\Big[r_{iC}(\varphi_i^d) + r_{iP}(\varphi_i^d)\Big] = w_i f^d, \qquad \frac{1}{\sigma}\, r_{iF}(\varphi_i^x) = w_i f^x. \tag{15}$$

Firms with $\varphi \in [\varphi_i^d, \varphi_i^x)$ serve only Home; firms with $\varphi \geq \varphi_i^x$ also export.

**Free entry.** Operating profits for a firm with productivity $\varphi$ in region $i$ are

$$\pi_i(\varphi) = \frac{1}{\sigma}\Big[r_{iC}(\varphi) + r_{iP}(\varphi) + \mathbf{1}\{\varphi \geq \varphi_i^x\}\, r_{iF}(\varphi)\Big] - w_i f^d - \mathbf{1}\{\varphi \geq \varphi_i^x\}\, w_i f^x. \tag{16}$$

Free entry requires that the expected value of entry equals the sunk cost:

$$w_i f^e = \int_{\varphi_i^d}^{\infty} \pi_i(\varphi)\, dG_i(\varphi). \tag{17}$$

This condition, together with the zero-profit conditions, pins down equilibrium entry and selection in each region. Let $M_i^e$ denote the mass of entrants and $M_i \equiv M_i^e[1 - G_i(\varphi_i^d)]$ the mass of active firms in region $i$.

**Price index.** Since manufacturing goods are freely traded within Home, all active firms serve both regions. The manufacturing price index is therefore common across regions:

$$P_C^M = P_P^M \equiv P^M = \left( \sum_{j \in \{C,P\}} M_j \int_{\varphi_j^d}^{\infty} p_j(\varphi)^{1-\sigma}\, \frac{dG_j(\varphi)}{1 - G_j(\varphi_j^d)} \right)^{\frac{1}{1-\sigma}}, \tag{18}$$

where the integral is over the conditional productivity distribution of active firms.

### 7.2.5 Labor Markets

Total Home population is fixed at $\bar{L}$. In each region, labor is allocated between agriculture and manufacturing. Labor market clearing therefore requires

$$\bar{L} = L_C + L_P = (L_C^A + L_C^M) + (L_P^A + L_P^M), \tag{19}$$

where agricultural labor demand is given by inverting the first-order condition (11):

$$L_i^A = \bar{T}_i \left( \frac{p_A \eta \, a_A}{w_i} \right)^{\frac{1}{1-\eta}}, \tag{20}$$

and manufacturing labor demand includes labor allocated to entry, fixed costs of domestic production, fixed export costs, and variable production:

$$L_i^M = M_i^e f^e + M_i \left[ \frac{1}{1 - G_i(\varphi_i^d)} \int_{\varphi_i^d}^{\infty} \ell_i(\varphi) \, dG_i(\varphi) + f^d + \left( \frac{\varphi_i^d}{\varphi_i^x} \right)^{\alpha_i} f^x \right], \tag{21}$$

where $\ell_i(\varphi)$ denotes variable labor input for a firm with productivity $\varphi$.

### 7.2.6 General Equilibrium

A general equilibrium consists of regional wages, population levels, manufacturing labor allocations, productivity cutoffs, and the mass of active firms, $\{w_i, L_i, L_i^M, \varphi_i^d, \varphi_i^x, M_i\}_{i \in \{C,P\}}$, together with a common manufacturing price index $P^M$, such that:

1. Workers choose locations to maximize utility, with the equilibrium population distribution satisfying (9).

2. Agricultural factor prices are determined by perfect competition (11), with wages equalizing across sectors within each region.

3. Manufacturing firms set prices to maximize operating profits, and the domestic and export productivity cutoffs in each region satisfy the zero-profit conditions (15).

4. Free entry drives expected profits of manufacturing firms in each region to zero (17).

5. The common manufacturing price index is given by (18).

6. Total population satisfies (19), and labor markets clear within each region, with manufacturing labor demand given by (21).

The equilibrium can be formulated as a fixed-point problem in $(w_C, w_P, L_C, L_P)$, with all remaining endogenous variables determined in closed form conditional on these four objects.[13]

## 7.3 Comparative Statics

In this model, a reduction in TPU is represented as a decline in the iceberg trade cost, $\tau$. A lower $\tau$ increases variable export revenues, reduces the export productivity cutoff, and expands export participation. The resulting rise in manufacturing labor demand induces spatial reallocation through endogenous migration.

Let $s_i \equiv L_i^M / L_i$ denote the manufacturing employment share in region $i$. Because $L_i^M = s_i L_i$, variation in manufacturing employment combines within-region structural transformation and shifts in the size of the regional labor force.

[13] The system of equilibrium conditions is solved numerically using MATLAB's `fsolve`. Closed-form expressions for the integrals in (17), (18), and (21), derived under Pareto distribution, are provided in Appendix B.2.

**Proposition 1 (Decomposition of Structural Change)** *The response of manufacturing employment to a change in the iceberg trade cost $\tau$ can be decomposed as:*

$$\frac{d \ln L_i^M}{d\tau} = \frac{1-s_i}{s_i} \cdot b \cdot \frac{d \ln w_i}{d\tau} + \frac{1}{s_i} \cdot \frac{d \ln L_i}{d\tau},$$

*where $b = 1/(1-\eta) > 1$ is the labor demand elasticity in agriculture.*

**Proof.** See Appendix B.1. ■

The decomposition reveals how a uniform reduction in variable trade costs can generate divergent manufacturing outcomes across regions through two distinct channels:

1. **Wage effect** $\left(\frac{1-s_i}{s_i} \cdot b \cdot \frac{d \ln w_i}{d\tau}\right)$: A reduction in $\tau$ increases export profits, bidding up local wages. Higher wages increase the marginal cost of agricultural production, inducing a reallocation of labor from agriculture to manufacturing. This channel operates in the same direction for both regions: a lower $\tau$ tends to expand manufacturing employment.

2. **Migration effect** $\left(\frac{1}{s_i} \cdot \frac{d \ln L_i}{d\tau}\right)$: Population inflows expand manufacturing employment, whereas outflows contract it. The amplification factor $1/s_i > 1$ implies that a one percent increase in regional population raises manufacturing employment by more than one percent, because agriculture faces diminishing returns due to fixed land and marginal workers are disproportionately absorbed by manufacturing. This channel operates asymmetrically across regions: if a lower $\tau$ draws workers toward Core and away from Periphery, it expands manufacturing in Core while contracting it in Periphery.

The key insight is that these two channels can operate in opposite directions for the Periphery. A decline in $\tau$ raises wages in both regions, so the wage effect increases $L_i^M$ everywhere. At

the same time, a rise in the Core's relative real income induces migration toward it, so the migration effect raises $L_C^M$ but reduces $L_P^M$. Whether manufacturing employment in the Periphery ultimately expands or contracts depends on the relative strength of these forces, a quantitative issue I investigate through the numerical simulations below.

## 7.4 Numerical Simulation

This subsection presents a numerical simulation to examine how a reduction in variable trade costs generates divergent manufacturing employment trajectories across Core and Periphery regions. The exercise serves two purposes: to confirm that the model's qualitative predictions hold within a plausible parameter range and are consistent with the empirical patterns documented in Section 5, and to quantify the relative contributions of the wage and migration channels identified in Proposition 1.

### 7.4.1 Parameterization

I parameterize the model using a combination of external estimates and internal calibration, as summarized in Table A18. The key estimated parameters are the Pareto shape parameters $\alpha_C = 3.730$ and $\alpha_P = 4.722$, which govern the thickness of the right tail of each region's productivity distribution.[14] A lower $\alpha$ implies a thicker right tail—i.e., a larger mass of highly productive firms—and therefore greater scope for export participation and gains from trade liberalization. The condition $\alpha_C < \alpha_P$ captures the empirical regularity that core regions host a disproportionate share of highly productive firms.

[14] I use estimates based on the top 10% of firms from Table 6. Alternative choices do not materially affect the results, provided that $\alpha_C < \alpha_P$ and $\alpha_C > \sigma - 1$, the latter being required for equilibrium to exist.

The elasticity of substitution across manufacturing varieties is set to $\sigma = 4$, following Broda and Weinstein (2006), and the agricultural labor share is $\eta = 0.5$, standard in the development literature (Restuccia et al., 2008). The fixed cost structure $(f^e, f^d, f^x) = (2, 3, 9)$ satisfies the theoretical ordering $f^e < f^d < f^x$ and ensures that only a subset of active firms select into exporting. While the absolute levels of these fixed costs are not pinned down by external evidence, the ratios are within ranges commonly assumed in the heterogeneous firm literature.

The agricultural productivity $a_A$ and the subsistence requirement $\bar{A}$ are both normalized to one. Given these normalizations, two parameters remain to be calibrated. The land endowments are set to $(\bar{T}_C, \bar{T}_P) = (30, 170)$, implying that core counties hold 15% of total agricultural land. By construction, core counties correspond to the top quartile of counties ranked by their pre-treatment manufacturing employment share. Many are urban districts with limited arable land, making the implied land distribution a plausible approximation. The foreign demand composite, $\Phi_F \equiv P_F^{\sigma-1} E_F$, is then calibrated to 62.4 so that the baseline equilibrium matches the population ratio $L_C/L_P = 0.67$ observed in the 2000 Population Census. The Core wage premium is governed by the difference in Pareto shape parameters, $\alpha_P - \alpha_C$, and is not directly targeted. Because these shape parameters are estimated from firm sales rather than from wages, the implied premium is an out-of-sample check: the calibrated model gives $w_C/w_P = 1.28$, against 1.29 in the ASIF. Agricultural goods are freely traded internationally at the common price $p_A$, allowing Home to import food to satisfy subsistence requirements. Trade is balanced in equilibrium, with the value of manufacturing exports to Foreign equal to the value of agricultural imports.

### 7.4.2 Diverging Effects of Trade Liberalization

The model is solved for a sequence of variable trade costs $\tau$ ranging from 1.4 to 1.05. Figure 6 displays the equilibrium wages, population, and manufacturing employment as functions of $\tau$.

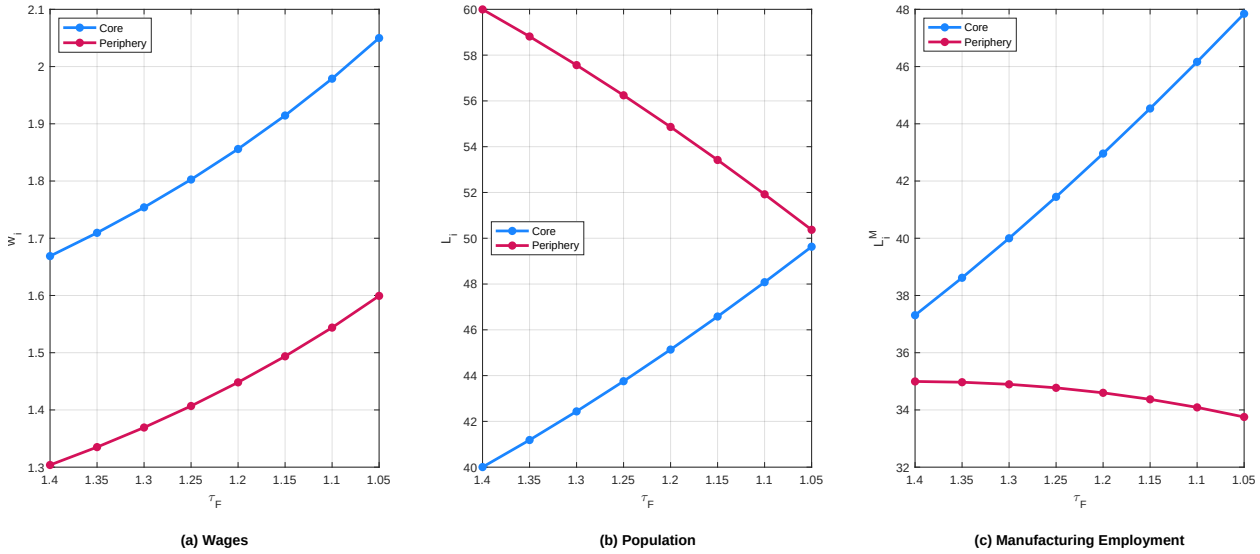



Figure 6: Trade Liberalization and Core–Periphery Divergence

*Notes:* This figure illustrates the model's responses to a reduction in variable trade costs $\tau$. Panel (a) reports equilibrium wages, panel (b) population, and panel (c) manufacturing employment in the Core and Periphery. Wages rise by similar proportions in both regions, leaving the Core wage premium approximately unchanged as trade costs fall. Population shifts toward the Core, causing manufacturing employment to expand in the Core and contract in the Periphery.

Panel (a) shows that wages rise in both regions as trade costs fall, reflecting the standard gains-from-trade effect: lower $\tau$ expands export opportunities, increases manufacturing revenue, and bids up the price of labor. However, wages in the Core remain consistently higher than in the Periphery, and the proportional gap is close to constant as trade liberalizes. When $\tau = 1.4$, the Core wage premium is about 28.0 percent; when $\tau = 1.05$, it increases to 28.2 percent. This wage premium arises from the thicker right tail of the Core's productivity distribution. Because $\alpha_C < \alpha_P$, the Core contains a larger share of highly productive firms, which generate higher revenues per worker in both domestic and export markets.

Panel (b) presents the corresponding population dynamics. The Core's share of total population increases from 40.0 percent to 49.6 percent as $\tau$ falls from 1.4 to 1.05, while the Periphery's share declines symmetrically. Migration responds to real income differentials, and since the manufacturing price index $P^M$ is common across regions, the relevant comparison is between disposable incomes per capita. At the baseline these are close together: the Core's wage advantage is offset by the rental income per resident that the Periphery's larger land endowment delivers. As trade costs fall, Core disposable income rises faster, and workers relocate until the population split is again consistent with the real income gap. The fixed supply of land damps this reallocation, since each arriving worker dilutes the rental income of Core residents, and the size of the resulting population shift underscores the strength of the export-driven pull when productivity tails differ substantially across regions.

Panel (c) reports the response of manufacturing employment, the central outcome of the model and the one most directly linked to the empirical evidence. Manufacturing employment in the Core rises substantially from 37.3 to 47.9, roughly matching the inflow of labor from the Periphery and indicating that migrants are absorbed primarily into the manufacturing sector. In contrast, manufacturing employment in the Periphery declines slightly from 35.0 to 33.8. This pattern of rising wages in both regions alongside manufacturing expansion in the Core alone mirrors the empirical evidence in Table 2, where the manufacturing gains from TPU reduction accrue disproportionately to counties with an established manufacturing base, relative to agricultural counties. Although the model includes only two regions, counties with greater exposure to the trade shock would move further along the corresponding trajectories, depending on whether their initial characteristics resemble the Core or the Periphery. More

exposed Core counties thus experience larger manufacturing expansions, while more exposed Periphery counties experience larger contractions.

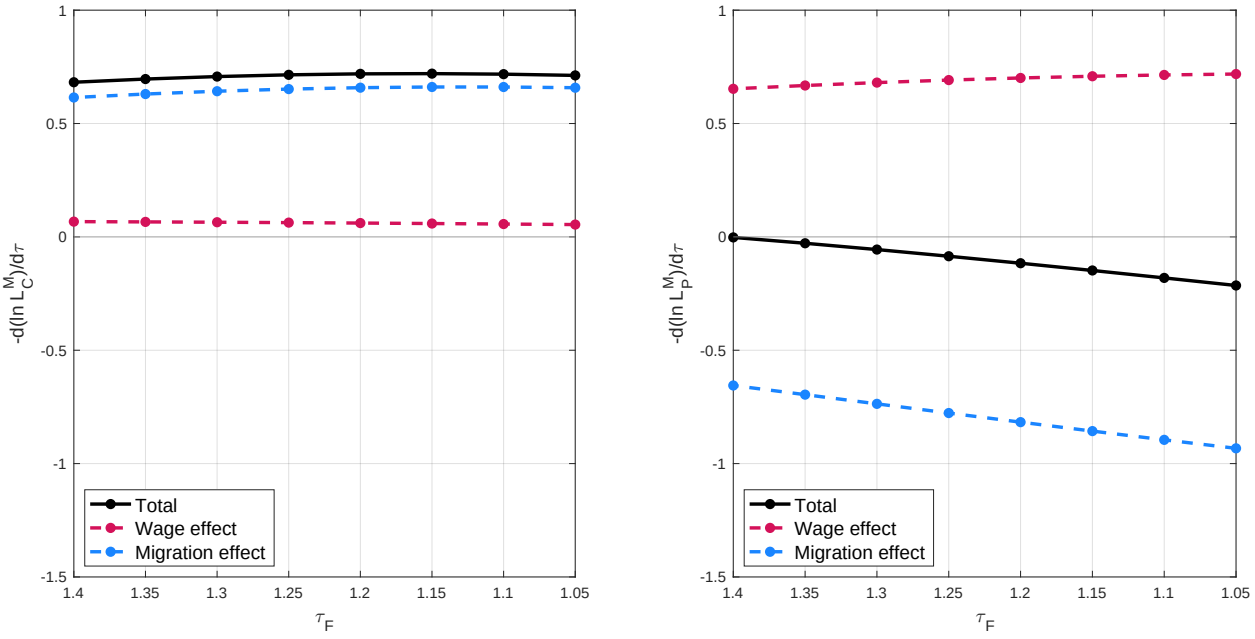
It is worth noting that the manufacturing employment *share* increases in both regions, because in the Periphery out-migration outpaces the contraction of the manufacturing sector. This is consistent with the empirical pattern in Table 5, where the Periphery's rising manufacturing share reflects a shrinking denominator rather than genuine industrial expansion.

### 7.4.3 Decomposition of Manufacturing Employment

To understand the forces behind this divergence, Figure 7 decomposes the semi-elasticity $\partial \ln L_i^M / \partial \tau$ into the wage and migration channels as in Proposition 1. For ease of interpretation, the sign is reversed so that positive values indicate that a reduction in $\tau$ increases $L_i^M$.

Panel (a) shows the decomposition for the Core. Both channels are positive, indicating that a reduction in $\tau$ unambiguously expands manufacturing employment. Lower trade costs raise wages, pulling labor out of agriculture and into manufacturing, while the Core simultaneously attracts workers from the Periphery. The wage effect is much smaller in magnitude, however, because the Core's initial manufacturing employment share is already high, leaving relatively little agricultural labor to reallocate. Therefore, the bulk of Core manufacturing growth is driven by population inflows rather than within-region structural change.

Panel (b) shows the decomposition for the Periphery, where the two channels work in *opposite* directions. The wage effect is positive (as in the Core): trade liberalization raises Periphery wages too, drawing workers from agriculture to manufacturing. However, the migration effect is negative and large: as workers leave the Periphery for the Core, the total labor pool shrinks, reducing manufacturing employment. The migration effect offsets the wage effect throughout




**(a) Core** **(b) Periphery**

Figure 7: Decomposition of Manufacturing Employment Responses

*Notes:* This figure decomposes the response of manufacturing employment $L_i^M$ to changes in trade costs $\tau_F$. The total response semi-elasticity $-d\ln L_i^M/d\tau$ is separated into a wage effect and a migration effect. The wage effect captures the direct impact of wage changes on sectoral labor demand, holding population fixed, while the migration effect reflects changes in regional population induced by real income differentials.

the range of $\tau$, so that the total semi-elasticity is negative across the range. At the $\tau = 1.4$ baseline the two channels almost exactly cancel, and the gap between them widens steadily as trade costs fall.

This decomposition reveals the model's central mechanism: trade liberalization generates a *within-region* force that expands manufacturing everywhere (the wage effect), but the *between-region* reallocation of labor (the migration effect) more than offsets this expansion in the Periphery while amplifying it in the Core. The asymmetry originates in the difference in Pareto shape parameters: with $\alpha_C < \alpha_P$, the Core has a thicker right tail, a lower export cutoff, and greater export revenue per unit of manufacturing labor. The Core thus benefits disproportionately when $\tau$ falls. Because labor is mobile, that advantage registers in quantities more than

in prices: workers relocate until real incomes are realigned, so the realized wage premium is almost unchanged while the Core's population share rises by roughly ten percentage points. The inflow expands the Core's manufacturing base and depletes the Periphery's workforce, which is exactly why the migration term dwarfs the wage term for the Core in Figure 7. The result is a divergence in manufacturing employment levels, even though wages rise in both regions.

The strength of this divergence depends critically on the migration elasticity $\nu$. As shown in Figure A7, when $\nu$ is low, workers are relatively insensitive to real income differentials, and the migration effect is weak. In this case, the wage effect dominates in both regions, and manufacturing employment expands everywhere. As $\nu$ increases past a threshold $\nu^*$, the migration response becomes strong enough to overwhelm the wage effect in the Periphery, tipping its total semi-elasticity from positive to negative. The baseline value $\nu = 5$ places the economy in this regime, consistent with the empirical finding that periphery counties experienced manufacturing decline following the reduction in TPU.

### 7.4.4 Discussion

The simulation results are qualitatively consistent with the empirical evidence on China's regional manufacturing divergence during the period of WTO accession and trade liberalization. The model generates three patterns that align with the data: (i) wages rise in all regions, (ii) population flows toward core regions, and (iii) manufacturing employment grows in the core while contracting slightly in the periphery.

The model does not incorporate several forces that are likely important in practice. Most notably, it abstracts from the rapid productivity growth that Chinese firms experienced during this period, which in the data delivered manufacturing expansion in most counties in absolute

terms. Therefore, the model's predictions should be interpreted as statements about *relative* divergence. In addition, the model takes the regional difference in Pareto shape parameters as given, without providing a micro-foundation for why core regions exhibit thicker-tailed productivity distributions. In practice, this difference likely reflects agglomeration externalities, such as knowledge spillovers, thick input markets, or better access to skilled labor. These same forces would also strengthen the migration effect beyond what the baseline calibration implies: if agglomeration externalities make the Core more attractive to workers, the effective migration response to trade liberalization would be larger at any given $\nu$, and the threshold $\nu^*$ above which the Periphery experiences relative manufacturing decline would be lower. The migration elasticity needed to generate divergence in this model should therefore be read as an upper bound on what a richer environment would require.

The results also carry policy implications. Spatial divergence in manufacturing employment can arise as a pure general equilibrium consequence of uniform trade liberalization, without any region-specific policy or differential market access. This suggests a rationale for place-based industrial policies, such as lowering barriers to firm entry or fostering technology adoption in lagging regions, that shift the local productivity distribution and strengthen the wage effect relative to the migration effect. At the same time, the welfare implications of such divergence are nuanced: in the model, real incomes rise in both regions as trade costs fall, so that workers in the Periphery benefit from incomes that rise faster than the manufacturing price index even as manufacturing employment declines. Policy discussions that equate manufacturing employment loss with welfare loss therefore risk overstating the costs borne by periphery regions.

# 8 Conclusion

This paper studies how the reduction in trade policy uncertainty following China's WTO accession reshaped the spatial distribution of manufacturing activity within China. I find sharply divergent local responses to the same shock: counties with established manufacturing bases expanded, while periphery counties experienced relative manufacturing decline and population outflows. A spatial equilibrium framework with heterogeneous firms and internal migration rationalizes these patterns through labor reallocation toward more productive regions. The results highlight an important distinction between the aggregate gains from expanded foreign market access and their spatial incidence: manufacturing expansion can be concentrated in already-industrialized regions, masking adverse effects elsewhere in aggregate statistics. In this sense, greater access to foreign markets can generate local losses through domestic spatial reallocation, even in the absence of increased import competition.

The external validity of these findings depends on the conditions underlying this spatial adjustment. The mechanism is likely to be more relevant in economies where manufacturing capacity is initially unevenly distributed across space, trade liberalization disproportionately expands opportunities in tradable industries, and workers are sufficiently mobile to respond to these opportunities. These features are not unique to China and characterize, to varying degrees, other export-oriented developing economies such as Vietnam and Bangladesh. At the same time, differences in internal migration frictions, industrial geography, and the sectoral composition of trade imply that the magnitude, and potentially even the direction, of spatial adjustment need not be the same across countries. Testing whether the spatial divergence documented here appears in other settings is left for future work.

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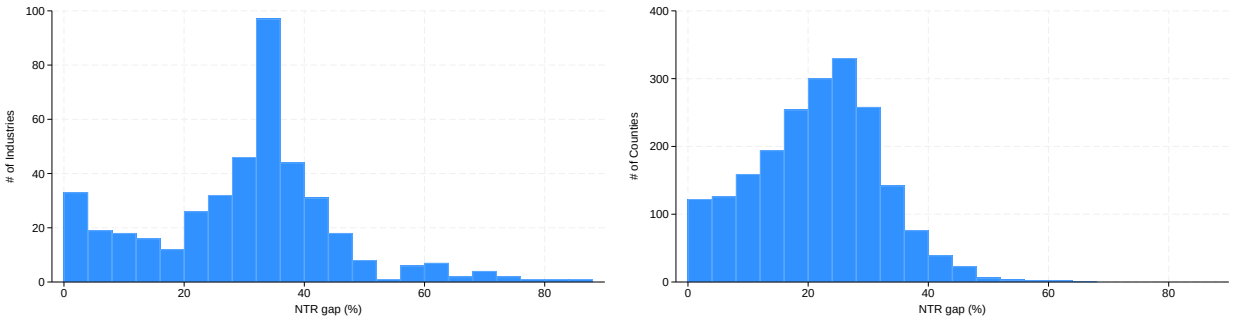
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# A Additional Tables and Figures




(a) Industries (b) Counties

Figure A1: Distribution of NTR Gaps

*Notes:* This figure plots the distribution of NTR gaps across 425 four-digit CIC manufacturing industries and 2,037 counties. County-level NTR gaps are constructed as the 1998 employment-share-weighted average of industry-level NTR gaps. The substantial cross-county variation provides the identifying variation for the shift-share design.

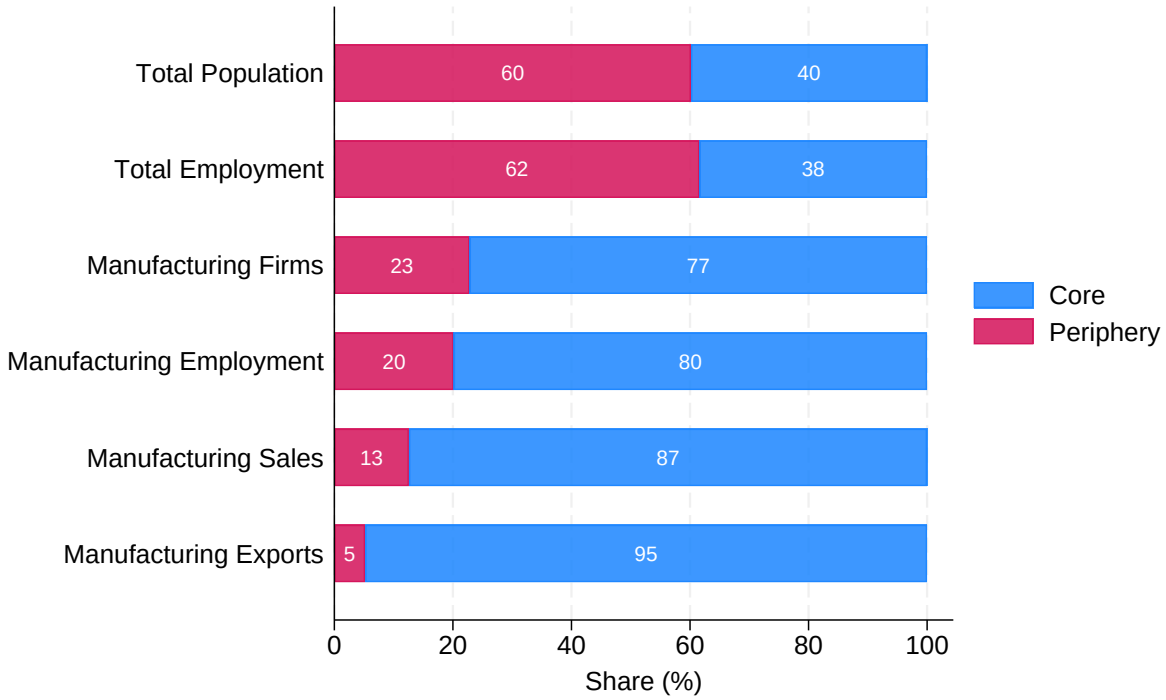
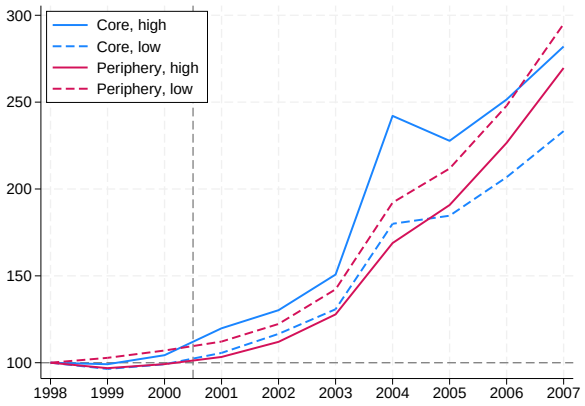
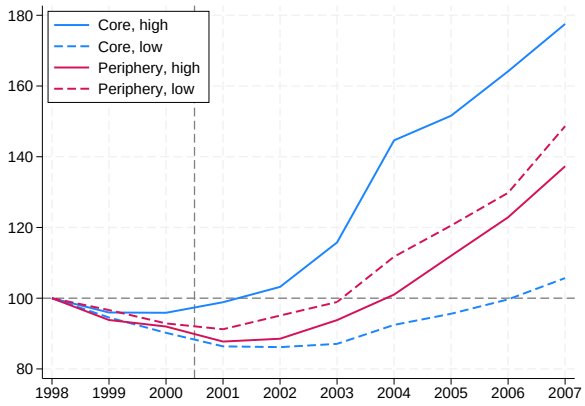



Figure A2: Core-Periphery Shares Across Key Outcome Variables

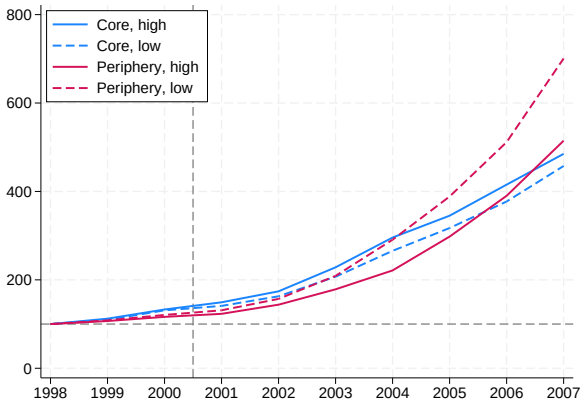
*Notes:* This figure reports the shares of core and periphery counties in total population, total employment, and a set of manufacturing outcomes. Counties are classified as core or periphery based on whether their manufacturing employment share exceeds the top-quartile threshold of 9.705 percent. Manufacturing outcomes include the number of manufacturing firms, manufacturing employment, manufacturing sales, and manufacturing exports. All bars sum to 100 percent.



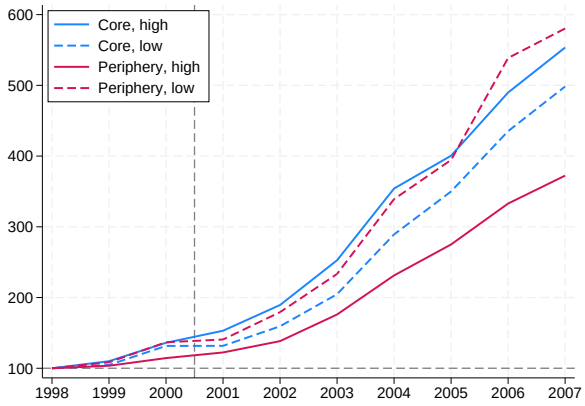

(a) Firms




(b) Employment




(c) Sales




(d) Exports

Figure A3: Manufacturing Activity in Absolute Terms, 1998–2007

*Notes:* This figure plots total manufacturing activity for each group of counties, indexed to 1998 = 100. Counties are ranked by their pre-treatment manufacturing employment share and divided into quartiles; the core consists of counties in the top quartile, and the periphery consists of the remaining counties. Within each group, counties are split at the median NTR gap into high- and low-exposure halves. Every group ends the period above its 1998 level. Within the core, the more-exposed half grows faster on every margin; within the periphery, it grows more slowly. Employment fell in all groups during the state-sector restructuring of the late 1990s and recovered thereafter.

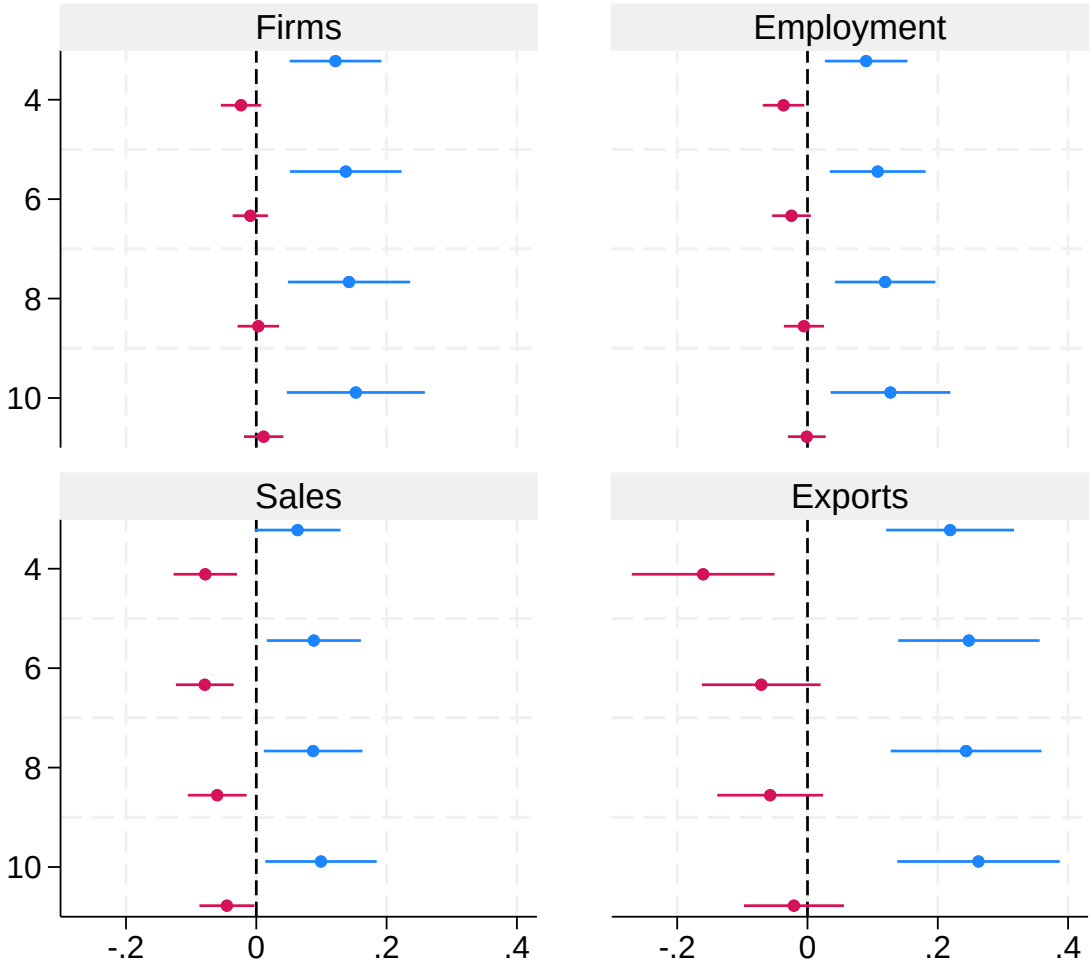



Figure A4: Sensitivity to Alternative Number of Quantiles

*Notes:* This figure presents the spatial reallocative effects of TPU reduction under alternative numbers of quantiles, showing the coefficient estimates and 95% confidence intervals. Numbers on the y-axis indicate the number of quantiles. Regressions are estimated separately for the top quantile (blue) and the remaining quantiles (red) using PPML. Standard errors are clustered at the county level.

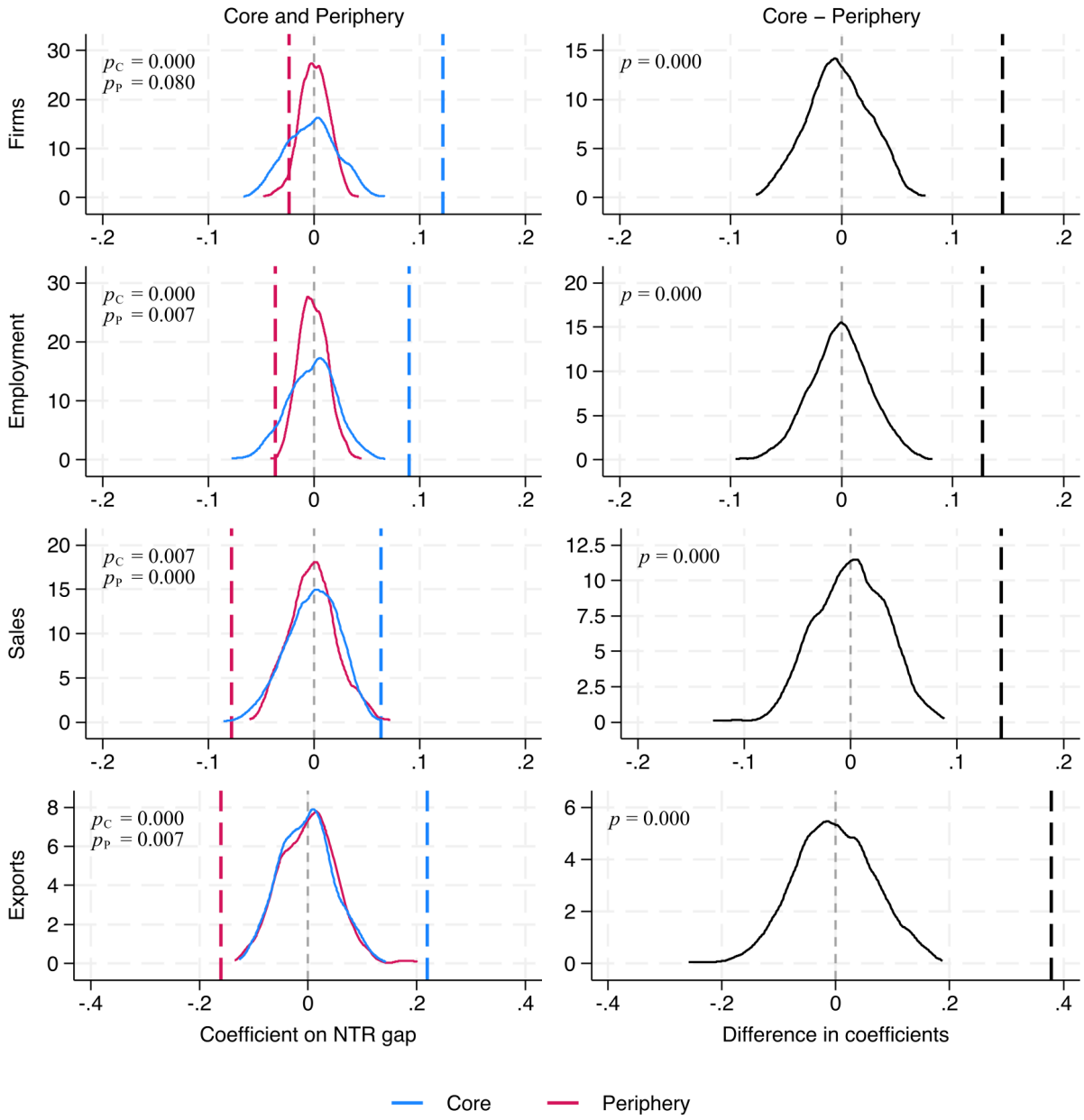



Figure A5: Placebo Tests: County-Level Permutations

*Notes:* This figure shows the distribution of placebo estimates of the coefficient on the NTR gap based on 300 random permutations of county-level exposure. In each permutation, NTR gaps are reassigned across the 2,037 counties, preserving their empirical distribution and holding all other county characteristics fixed. The left column plots kernel density estimates of the placebo coefficients separately for core and periphery counties, while the right column reports the core–periphery difference. Dashed lines mark the baseline estimates obtained under the true assignment. Reported $p$-values are two-sided placebo $p$-values, defined as the share of permutations yielding estimates at least as large in absolute value as the baseline estimate. All regressions use the same PPML specification as in the baseline analysis, with standard errors clustered at the county level.

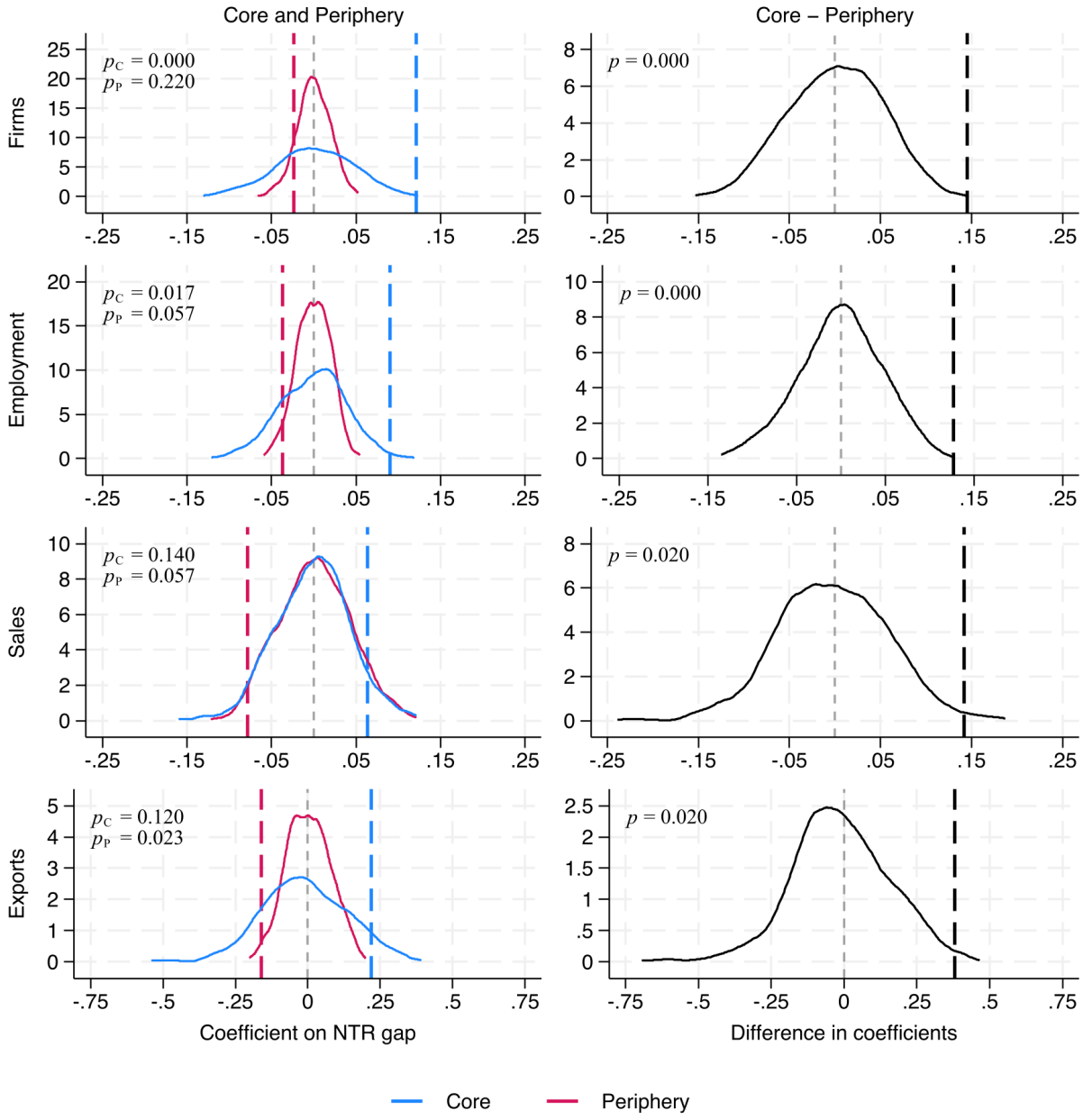
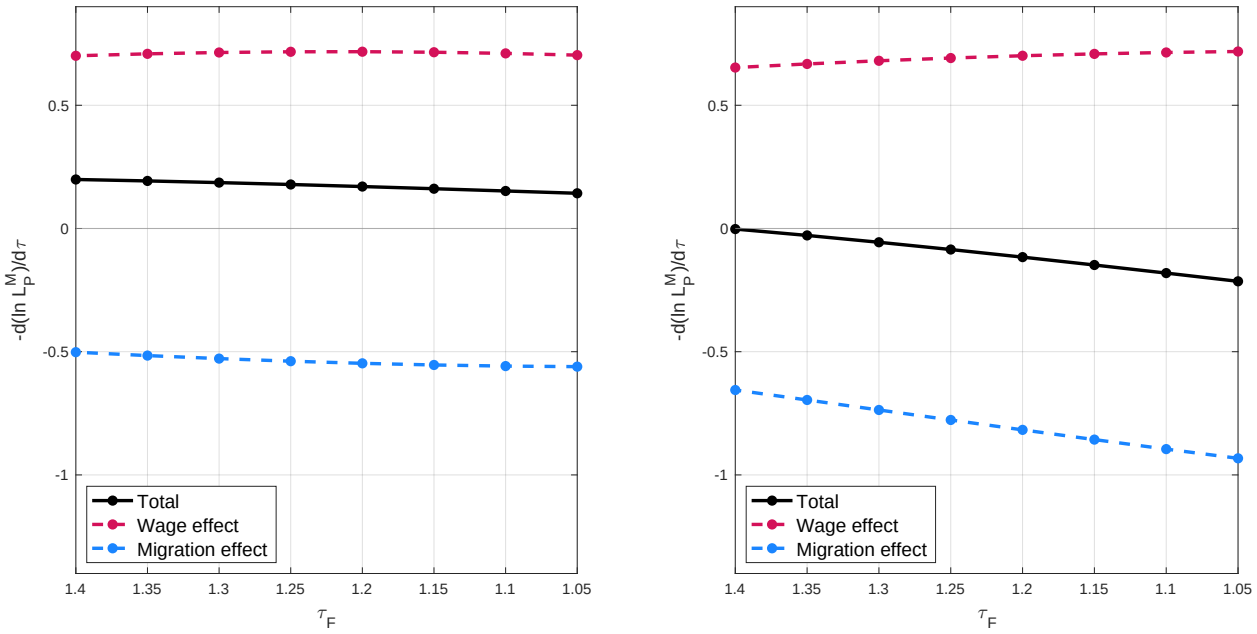



Figure A6: Placebo Tests: Industry-Level Permutations

*Notes:* This figure shows the distribution of placebo estimates of the coefficient on the NTR gap based on 300 random permutations of the underlying industry shocks. In each permutation, NTR gaps are reassigned across four-digit CIC industries, preserving their empirical distribution, and county-level exposure is rebuilt from the permuted industry gaps using each county's true 1998 employment shares. The left column plots kernel density estimates of the placebo coefficients separately for core and periphery counties, while the right column reports the core–periphery difference. Dashed lines mark the baseline estimates obtained under the true assignment. Reported $p$-values are two-sided placebo $p$-values, defined as the share of permutations yielding estimates at least as large in absolute value as the baseline estimate. All regressions use the same PPML specification as in the baseline analysis, with standard errors clustered at the county level.




**(a) Periphery, $\nu = 2$** **(b) Periphery, $\nu = 5$**

Figure A7: Role of Migration Elasticity

*Notes:* This figure decomposes the response of Periphery manufacturing employment $L_P^M$ to changes in trade costs $\tau_F$ under two values of the migration elasticity $\nu$. The total response semi-elasticity $-d \ln L_P^M / d\tau$ is separated into a wage effect and a migration effect. The wage effect captures the direct impact of wage changes on sectoral labor demand, holding population fixed, while the migration effect reflects changes in regional population induced by real income differentials. When $\nu$ is low, the wage effect dominates and trade liberalization expands Periphery manufacturing. At higher $\nu$, the migration effect overwhelms the wage effect, turning the total response negative.

Table A1: Summary Statistics of Variables

| | Obs. | Non-zeros | Mean | S.D. | Median | Min | Max |
|---|---|---|---|---|---|---|---|
| **Panel A. Manufacturing outcomes** | | | | | | | |
| Firms | 20,370 | 20,044 | 87.81 | 384.12 | 20 | 0 | 15016 |
| Employment | 20,370 | 20,044 | 24.63 | 97.98 | 5.26 | 0 | 2998.74 |
| Sales | 20,370 | 20,044 | 6.19 | 35.65 | 0.66 | 0 | 1660.80 |
| Exports | 20,370 | 13,060 | 1.33 | 12.78 | 0.02 | 0 | 636.14 |
| **Panel B. Trade policy uncertainty** | | | | | | | |
| NTR gap | 2,037 | 2,033 | 21.61 | 10.45 | 22.29 | 0 | 64.06 |
| NTR gap (normalized) | 2,037 | 2,037 | 0 | 1 | 0.07 | -2.07 | 4.06 |
| **Panel C. Time-varying tariff controls** | | | | | | | |
| China import tariff | 20,370 | 20,335 | 15.04 | 10.34 | 12.35 | 0 | 106.58 |
| US applied tariff | 20,370 | 19,567 | 2.93 | 2.11 | 2.63 | 0 | 21.15 |
| EU applied tariff | 20,370 | 19,276 | 2.51 | 1.80 | 2.35 | 0 | 16.57 |
| Japan applied tariff | 20,370 | 17,705 | 1.67 | 1.99 | 1.14 | 0 | 20.48 |
| S. Korea applied tariff | 20,370 | 20,334 | 8.73 | 4.79 | 7.83 | 0 | 72.71 |
| **Panel D. Initial county characteristics** | | | | | | | |
| Initial manu. emp. share | 2,037 | 2,037 | 8.36 | 9.15 | 5.11 | 0.14 | 75.56 |
| Initial SOE share | 2,037 | 1,938 | 44.10 | 29.21 | 40.91 | 0 | 100 |
| Distance to port | 2,037 | 2,037 | 242.55 | 459.47 | 103.68 | 0.003 | 3078.59 |
| Initial pop. density | 2,037 | 2,037 | 365.03 | 394.04 | 241.44 | 0.23 | 5867.04 |

*Notes:* This table presents summary statistics for the county-level panel from 1998 to 2007. The corresponding units are number for firms; thousands for employment; billions of RMB (1998 constant prices) for sales and exports; percent for NTR gaps, tariff rates, and shares; kilometer for distances; and people per square kilometer for population density. Panels A and C report statistics for the full panel ($N = 20,370$), while Panels B and D report cross-sectional statistics for the base year ($N = 2,037$).

Table A2: Ranking of NTR Gaps by Industry

| Rank | NTR gap (%) | CIC | CIC description | Emp. share (%) |
|---|---|---|---|---|
| 1 | 82.58 | 4216 | Drawnwork and embroidered handicrafts | 0.18 |
| 2 | 76.29 | 3060 | Plastic packing boxes and containers | 0.14 |
| 3 | 73.09 | 1743 | Colored or patterned silk fabrics | 0.17 |
| 4 | 70.86 | 1742 | Raw silk fabrics | 0.84 |
| 5 | 69.47 | 2440 | Toys | 0.81 |
| 6 | 64.15 | 3070 | Plastic parts | 0.11 |
| 7 | 61.37 | 3090 | Other plastic products | 0.45 |
| 8 | 60.61 | 1761 | Cotton and chemical-fibre knitted goods | 0.53 |
| 9 | 60.45 | 1810 | Textile apparel | 3.62 |
| 10 | 59.10 | 1942 | Feather and down products | 0.12 |
| | | | ... | |
| 210 | 2.15 | 1492 | Frozen drinks and edible ice | 0.11 |
| 211 | 1.71 | 2511 | Petroleum products | 1.34 |
| 212 | 0.54 | 3671 | Tractors | 0.41 |
| 213 | 0.41 | 1340 | Sugar refining | 0.60 |
| 214 | 0.32 | 1361 | Frozen aquatic products | 0.25 |
| 215 | 0.20 | 2311 | Books, newspapers, and periodicals | 0.44 |
| 216 | 0.06 | 2621 | Nitrogen fertilizer | 1.74 |
| 217 | 0.00 | 2622 | Phosphate fertilizer | 0.55 |
| 217 | 0.00 | 2520 | Coke oven products | 0.36 |
| 217 | 0.00 | 2624 | Compound fertilizer | 0.15 |

*Notes:* This table ranks four-digit CIC manufacturing industries by their NTR gap among the 219 industries that each account for at least 0.1 percent of manufacturing employment in 1998. Across all 426 manufacturing industries, the NTR gap ranges from zero, for thirteen industries, to 86.3 percent. The maximum belongs to oral hygiene products (2673), an industry accounting for only 0.04 percent of manufacturing employment in 1998.

Table A3: Classification of Manufacturing Industries

| Category | CIC | Industries |
|---|---|---|
| Food Processing | 13–16 | Farm and Sideline Food Products; Food Products; Beverages; Tobacco Products |
| Light Manufacturing | 17–24 | Textiles; Textile Wearing Apparel, Footwear, and Caps; Leather, Fur, Feather, and Related Products; Timber, Wood, Bamboo, Rattan, Palm and Straw Products; Furniture; Paper and Paper Products; Printing and Reproduction of Recording Media; Articles for Culture, Education, and Sports Activities |
| Heavy Manufacturing | 25–34 | Petroleum, Coking, and Nuclear Fuel Processing; Raw Chemical Materials and Chemical Products; Medicines; Chemical Fibers; Rubber Products; Plastic Products; Non-metallic Mineral Products; Ferrous Metal Smelting and Pressing; Non-ferrous Metal Smelting and Pressing; Metal Products |
| Machinery | 35–41 | General-Purpose Machinery; Special-Purpose Machinery; Transport Equipment; Electrical Machinery and Apparatus; Communication Equipment, Computers, and Other Electronics; Instruments, Meters, and Cultural and Office Machinery |

*Notes:* This table presents the classification of two-digit CIC manufacturing industries into four broad categories used in the analysis of structural change within manufacturing. CIC 42 (Crafts and Other Manufacturing) and CIC 43 (Recycling of Waste Resources and Materials) are excluded from the classification.

Table A4: Effects of TPU Reduction: Core Counties

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A. Firms** | | | | | |
| NTR gap × Post | 0.188*** | 0.123*** | 0.120*** | 0.126*** | 0.122*** |
| | (0.0372) | (0.0345) | (0.0354) | (0.0356) | (0.0359) |
| **Panel B. Employment** | | | | | |
| NTR gap × Post | 0.251*** | 0.113*** | 0.107*** | 0.103*** | 0.0900*** |
| | (0.0481) | (0.0324) | (0.0319) | (0.0317) | (0.0323) |
| **Panel C. Sales** | | | | | |
| NTR gap × Post | 0.0994*** | 0.0723** | 0.0698** | 0.0704** | 0.0634* |
| | (0.0354) | (0.0317) | (0.0322) | (0.0320) | (0.0336) |
| **Panel D. Exports** | | | | | |
| NTR gap × Post | 0.256*** | 0.213*** | 0.218*** | 0.225*** | 0.219*** |
| | (0.0533) | (0.0510) | (0.0502) | (0.0501) | (0.0501) |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | ✓ | ✓ | ✓ | ✓ | ✓ |
| Trends: Province | | ✓ | ✓ | ✓ | ✓ |
| Trends: Admin Status | | | ✓ | ✓ | ✓ |
| Trends: Port Quartile | | | | ✓ | ✓ |
| Trends: SOE Quartile | | | | | ✓ |
| # of Obs. (Panel A-C) | 5,220 | 5,220 | 5,220 | 5,220 | 5,220 |
| # of Obs. (Panel D) | 5,160 | 5,170 | 5,170 | 5,170 | 5,170 |

*Notes:* This table presents the effects of TPU reduction on core counties, defined as those in the top quartile of the pre-treatment manufacturing employment share. Column (1) includes only two-way fixed effects and tariff controls in addition to the DID term; columns (2) to (5) sequentially introduce interactions between county characteristics and time trends, with column (5) corresponding to the baseline specification. Each cell corresponds to a separate regression, with the dependent variable indicated in the panel header. All specifications are estimated using PPML, with standard errors clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Effects of TPU Reduction: Periphery Counties

| | | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|---|
| **Panel A. Firms** | | | | | | |
| NTR gap × Post | | -0.0457* | -0.0518*** | -0.0436*** | -0.0445*** | -0.0234 |
| | | (0.0236) | (0.0164) | (0.0163) | (0.0164) | (0.0158) |
| **Panel B. Employment** | | | | | | |
| NTR gap × Post | | -0.0301 | -0.0465*** | -0.0283* | -0.0262 | -0.0368** |
| | | (0.0194) | (0.0173) | (0.0165) | (0.0164) | (0.0163) |
| **Panel C. Sales** | | | | | | |
| NTR gap × Post | | -0.129*** | -0.108*** | -0.0924*** | -0.0897*** | -0.0783*** |
| | | (0.0325) | (0.0263) | (0.0263) | (0.0261) | (0.0248) |
| **Panel D. Exports** | | | | | | |
| NTR gap × Post | | -0.227*** | -0.173*** | -0.172*** | -0.176*** | -0.160*** |
| | | (0.0560) | (0.0525) | (0.0529) | (0.0534) | (0.0559) |
| County FE | | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year FE | | ✓ | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | | ✓ | ✓ | ✓ | ✓ | ✓ |
| Trends | Province | | ✓ | ✓ | ✓ | ✓ |
| | Admin Status | | | ✓ | ✓ | ✓ |
| | Port Quartile | | | | ✓ | ✓ |
| | SOE Quartile | | | | | ✓ |
| # of Obs. (Panel A-C) | | 15,150 | 15,150 | 15,150 | 15,150 | 15,150 |
| # of Obs. (Panel D) | | 12,050 | 12,050 | 12,050 | 12,050 | 12,050 |

*Notes:* This table presents the effects of TPU reduction on periphery counties, defined as those outside the top quartile of the pre-treatment manufacturing employment share. Column (1) includes only two-way fixed effects and tariff controls in addition to the DID term; columns (2) to (5) sequentially introduce interactions between county characteristics and time trends, with column (5) corresponding to the baseline specification. Each cell corresponds to a separate regression, with the dependent variable indicated in the panel header. All specifications are estimated using PPML, with standard errors clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Core–Periphery Divergence Across Specifications

| | | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|---|
| Firms | | 0.234*** | 0.175*** | 0.164*** | 0.170*** | 0.145*** |
| | | (0.0441) | (0.0382) | (0.0389) | (0.0392) | (0.0392) |
| Employment | | 0.281*** | 0.159*** | 0.135*** | 0.129*** | 0.127*** |
| | | (0.0519) | (0.0367) | (0.0360) | (0.0357) | (0.0362) |
| Sales | | 0.228*** | 0.180*** | 0.162*** | 0.160*** | 0.142*** |
| | | (0.0480) | (0.0412) | (0.0416) | (0.0413) | (0.0418) |
| Exports | | 0.483*** | 0.386*** | 0.390*** | 0.400*** | 0.379*** |
| | | (0.0773) | (0.0732) | (0.0729) | (0.0732) | (0.0751) |
| County FE | | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year FE | | ✓ | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | | ✓ | ✓ | ✓ | ✓ | ✓ |
| Trends | Province | | ✓ | ✓ | ✓ | ✓ |
| | Admin Status | | | ✓ | ✓ | ✓ |
| | Port Quartile | | | | ✓ | ✓ |
| | SOE Quartile | | | | | ✓ |

*Notes:* This table reports the extent of core–periphery divergence across specifications. Each cell reports the difference between the estimated effects for core and periphery counties under the specification indicated by the column. The underlying estimates are reported in Tables A4 and A5. Standard errors of the differences are computed as the square root of the sum of the squared standard errors, since the core and periphery samples are disjoint. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Effects of TPU Reduction: Pooled Triple Interaction

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Firms | Employment | Sales | Exports |
| NTR gap × Post | -0.0499** | -0.0827*** | -0.136*** | -0.121** |
| | (0.0218) | (0.0203) | (0.0341) | (0.0546) |
| NTR gap × Post × Core | 0.187*** | 0.188*** | 0.232*** | 0.321*** |
| | (0.0351) | (0.0309) | (0.0440) | (0.0594) |
| County FE | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | ✓ | ✓ | ✓ | ✓ |
| Trends | ✓ | ✓ | ✓ | ✓ |
| # of Obs. | 20,370 | 20,370 | 20,370 | 17,220 |

*Notes:* This table presents heterogeneous effects of TPU reduction on manufacturing outcomes from a pooled specification that includes an interaction between county exposure and an indicator for core counties. All specifications are estimated by PPML, with standard errors clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Pre-Trends and Core–Periphery Divergence in the Event Study

| | (1) Firms | (2) Employment | (3) Sales | (4) Exports |
|---|---|---|---|---|
| **Panel A. Difference in event-study coefficients (Core − Periphery)** | | | | |
| 1998 | -0.00973 | 0.0394 | 0.0407 | -0.0402 |
| | (0.0317) | (0.0254) | (0.0320) | (0.0616) |
| 1999 | -0.00880 | 0.0113 | 0.0422* | -0.0167 |
| | (0.0172) | (0.0172) | (0.0221) | (0.0433) |
| 2001 | 0.0555*** | 0.0406* | 0.0411* | 0.193*** |
| | (0.0174) | (0.0219) | (0.0223) | (0.0527) |
| 2002 | 0.0698*** | 0.0637** | 0.0778*** | 0.296*** |
| | (0.0250) | (0.0266) | (0.0272) | (0.0686) |
| 2003 | 0.118*** | 0.128*** | 0.158*** | 0.349*** |
| | (0.0336) | (0.0298) | (0.0383) | (0.0792) |
| 2004 | 0.255*** | 0.271*** | 0.253*** | 0.511*** |
| | (0.0489) | (0.0457) | (0.0512) | (0.105) |
| 2005 | 0.175*** | 0.227*** | 0.228*** | 0.374*** |
| | (0.0461) | (0.0484) | (0.0536) | (0.0984) |
| 2006 | 0.166*** | 0.214*** | 0.209*** | 0.466*** |
| | (0.0491) | (0.0554) | (0.0607) | (0.120) |
| 2007 | 0.166*** | 0.205*** | 0.195*** | 0.409*** |
| | (0.0576) | (0.0625) | (0.0638) | (0.129) |
| **Panel B. Wald tests ($p$-values)** | | | | |
| *Group-specific pre-treatment coefficients jointly zero* | | | | |
| Core | 0.376 | 0.079 | 0.071 | 0.426 |
| Periphery | 0.111 | 0.705 | 0.263 | 0.403 |
| *Core–periphery differences jointly zero* | | | | |
| Pre-treatment | 0.875 | 0.240 | 0.150 | 0.808 |
| Post-treatment | <0.001 | <0.001 | <0.001 | <0.001 |

*Notes:* This table reports formal tests for the event-study estimates in Figure 4, estimated separately for core and periphery counties using equation (5) with 2000 as the omitted base year. Panel A reports annual core–periphery coefficient differences, with standard errors in parentheses. Positive values indicate larger exposure responses in the core. Panel B reports Wald tests of jointly zero group-specific pre-treatment coefficients and jointly zero core–periphery differences before and after treatment. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Sensitivity to Interaction Fixed Effects

| | (1) Firms | (2) Employment | (3) Sales | (4) Exports |
|---|---|---|---|---|
| **Quartiles of Manufacturing Employment Share** | | | | |
| **Panel A. Core** | | | | |
| NTR gap × Post | 0.0946** | 0.113*** | 0.0495 | 0.196*** |
| | (0.0425) | (0.0416) | (0.0393) | (0.0700) |
| Mean (2000) | 167.9 | 64.14 | 11.56 | 2.48 |
| # of Obs. | 5,210 | 5,210 | 5,210 | 5,160 |
| **Panel B. Periphery** | | | | |
| NTR gap × Post | -0.0208 | -0.0373** | -0.0932*** | -0.202*** |
| | (0.0171) | (0.0174) | (0.0275) | (0.0628) |
| Mean (2000) | 16.99 | 5.54 | 0.57 | 0.05 |
| # of Obs. | 15,150 | 15,150 | 15,150 | 12,044 |
| County FE | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | ✓ | ✓ | ✓ | ✓ |
| Interaction FEs | ✓ | ✓ | ✓ | ✓ |

*Notes:* This table presents robustness check results assessing the sensitivity to using interaction fixed effects instead of time trends. Counties are grouped into quartiles based on their pre-treatment manufacturing employment share, and regressions are estimated separately for the top quartile and the remaining quartiles using PPML. Standard errors are clustered at the county level. Stars denote significance levels as follows: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Sensitivity to Exclusion of Counties with Extreme NTR Gaps

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Firms | | | Employment | | |
| | Drop 1% | Drop 5% | Drop 10% | Drop 1% | Drop 5% | Drop 10% |
| **Quartiles of Manufacturing Employment Share** | | | | | | |
| **Panel A. Core** | | | | | | |
| NTR gap × Post | 0.149*** | 0.187*** | 0.200*** | 0.111*** | 0.146*** | 0.128*** |
| | (0.0321) | (0.0380) | (0.0461) | (0.0310) | (0.0354) | (0.0375) |
| # of Obs. | 5,110 | 4,690 | 4,170 | 5,110 | 4,690 | 4,170 |
| **Panel B. Periphery** | | | | | | |
| NTR gap × Post | -0.0306* | -0.0298 | -0.0182 | -0.0384** | -0.0318* | -0.0218 |
| | (0.0167) | (0.0189) | (0.0200) | (0.0171) | (0.0177) | (0.0220) |
| # of Obs. | 14,840 | 13,380 | 12,120 | 14,840 | 13,380 | 12,120 |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Trends | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

| | (7) | (8) | (9) | (10) | (11) | (12) |
|---|---|---|---|---|---|---|
| | Sales | | | Exports | | |
| | Drop 1% | Drop 5% | Drop 10% | Drop 1% | Drop 5% | Drop 10% |
| **Quartiles of Manufacturing Employment Share** | | | | | | |
| **Panel A. Core** | | | | | | |
| NTR gap × Post | 0.0808** | 0.0983** | 0.0775* | 0.228*** | 0.240*** | 0.256*** |
| | (0.0333) | (0.0413) | (0.0436) | (0.0534) | (0.0627) | (0.0687) |
| # of Obs. | 5,110 | 4,690 | 4,170 | 5,070 | 4,670 | 4,150 |
| **Panel B. Periphery** | | | | | | |
| NTR gap × Post | -0.0851*** | -0.109*** | -0.114*** | -0.182*** | -0.183** | -0.149* |
| | (0.0266) | (0.0328) | (0.0398) | (0.0577) | (0.0723) | (0.0843) |
| # of Obs. | 14,840 | 13,380 | 12,120 | 11,850 | 11,060 | 10,110 |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Trends | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes:* This table presents robustness check results assessing the sensitivity to excluding counties with extreme exposures to TPU reduction. Counties are grouped into quartiles based on their pre-treatment manufacturing employment share, and regressions are estimated separately for the top quartile and the remaining quartiles using PPML. Standard errors are clustered at the county level. Stars denote significance levels as follows: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Sensitivity to Alternative Estimation Methods

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | | Firms | | | Employment | |
| | OLS | WLS | WLS | OLS | WLS | WLS |
| | $\log(Y)$ | $\log(Y)$ | $\log(c+Y)$ | $\log(Y)$ | $\log(Y)$ | $\log(c+Y)$ |
| **Quartiles of Manufacturing Employment Share** | | | | | | |
| **Panel A. Core** | | | | | | |
| NTR gap × Post | 0.0844*** | 0.0750*** | 0.0760*** | 0.0742*** | 0.0753** | 0.0772** |
| | (0.0190) | (0.0289) | (0.0289) | (0.0204) | (0.0297) | (0.0301) |
| # of Obs. | 5,199 | 5,199 | 5,220 | 5,199 | 5,199 | 5,220 |
| **Panel B. Periphery** | | | | | | |
| NTR gap × Post | -0.0128 | -0.0326** | -0.0304** | -0.0267* | -0.0372** | -0.0370** |
| | (0.0127) | (0.0141) | (0.0137) | (0.0151) | (0.0157) | (0.0171) |
| # of Obs. | 14,844 | 14,844 | 15,150 | 14,844 | 14,844 | 15,150 |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Trends | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

| | (7) | (8) | (9) | (10) | (11) | (12) |
|---|---|---|---|---|---|---|
| | | Sales | | | Exports | |
| | OLS | WLS | WLS | OLS | WLS | WLS |
| | $\log(Y)$ | $\log(Y)$ | $\log(c+Y)$ | $\log(Y)$ | $\log(Y)$ | $\log(c+Y)$ |
| **Quartiles of Manufacturing Employment Share** | | | | | | |
| **Panel A. Core** | | | | | | |
| NTR gap × Post | 0.0375 | 0.0335 | 0.0498* | 0.136** | 0.178*** | 0.125 |
| | (0.0247) | (0.0242) | (0.0292) | (0.0599) | (0.0665) | (0.0850) |
| # of Obs. | 5,199 | 5,199 | 5,220 | 4,913 | 4,913 | 5,220 |
| **Panel B. Periphery** | | | | | | |
| NTR gap × Post | -0.0194 | -0.0611*** | -0.0611*** | -0.118** | -0.176*** | -0.456*** |
| | (0.0175) | (0.0186) | (0.0198) | (0.0481) | (0.0612) | (0.122) |
| # of Obs. | 14,844 | 14,844 | 15,150 | 8,018 | 8,018 | 15,150 |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Trends | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes:* This table presents robustness check results assessing the sensitivity to using log specification and alternative estimation methods. Counties are grouped into quartiles based on their pre-treatment manufacturing employment share, and regressions are estimated separately for the top quartile and the remaining quartiles using OLS and WLS. Standard errors are clustered at the county level. Stars denote significance levels as follows: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: Identifying Variation in the Shift-Share Design

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Within manufacturing | | | Incl. non-manufacturing | | |
| | All | Core | Periphery | All | Core | Periphery |
| **Panel A. Distribution of shocks, importance-weighted** | | | | | | |
| Mean (%) | 21.61 | 27.25 | 19.66 | 2.17 | 5.92 | 0.88 |
| S.D. (%) | 17.07 | 17.55 | 16.46 | 8.85 | 14.21 | 5.40 |
| Interquartile range (%) | 28.41 | 23.12 | 28.95 | 0.00 | 0.00 | 0.00 |
| Number of counties | 2,037 | 522 | 1,515 | 2,037 | 522 | 1,515 |
| **Panel B. Effective shock variation** | | | | | | |
| *Four-digit industries* | | | | | | |
| Number of shocks | 425 | 425 | 403 | 426 | 426 | 404 |
| 1/HHI | 42.8 | 80.0 | 32.6 | 1.2 | 1.6 | 1.1 |
| Largest weight | 0.114 | 0.057 | 0.136 | 0.916 | 0.796 | 0.958 |
| *Two-digit industries* | | | | | | |
| Number of shocks | 29 | 29 | 29 | 30 | 30 | 30 |
| 1/HHI | 13.3 | 17.5 | 11.5 | 1.2 | 1.6 | 1.1 |
| Largest weight | 0.166 | 0.122 | 0.188 | 0.916 | 0.796 | 0.958 |

*Notes:* This table reports the distribution of county exposure and the effective variation in the underlying industry shocks. Columns (1)–(3) use the exposure measure in equation (2), for which manufacturing employment shares sum to one within each county. Columns (4)–(6) rescale these shares by the county's 2000 manufacturing employment share and assign the remaining share to a non-manufacturing sector with a zero shock. Panel A reports the distribution of the industry-level NTR gaps weighted by their importance weights $s_n$, and Panel B reports the number of shocks, the inverse Herfindahl index of shock weights, and the largest shock weight at the four- and two-digit CIC levels. All statistics are computed using the `ssaggregate` routine of Borusyak et al. (2022), with moments weighted by exposure importance.

Table A13: Variance Decomposition of Industry NTR Gaps

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Variance | | Share of total | |
| | Est. | S.E. | Est. | S.E. |
| Two-digit component | 0.0087 | (0.0030) | 0.334 | (0.090) |
| Three-digit component | 0.0061 | (0.0020) | 0.234 | (0.077) |
| Four-digit residual | 0.0113 | (0.0024) | 0.432 | (0.065) |
| Total | 0.0261 | (0.0039) | 1.000 | — |

*Notes:* This table decomposes the variance of NTR gaps across the 425 four-digit CIC industries in 1998 using a nested random-effects model, with three-digit industries nested within two-digit industries. The two-digit component captures variation common to four-digit industries within the same two-digit industry; the three-digit component captures additional variation common to four-digit industries within the same three-digit industry; and the residual captures four-digit-industry-specific variation. Standard errors are computed using the delta method. Variance shares sum to one by construction.

Table A14: Exposure-Robust Inference

| | | (1) | (2) | (3) | (4) |
|---|---|---|---|---|---|
| | | | Standard error | | |
| Outcome | | Coef. | County cluster | Four-digit industry | Two-digit industry |
| Firms | Core | 0.122 | (0.036) *** | (0.034) *** | (0.038) *** |
| | Periphery | -0.023 | (0.016) | (0.024) | (0.026) |
| | Core − Periphery | 0.145 | (0.039) *** | (0.046) *** | (0.051) *** |
| Employment | Core | 0.090 | (0.032) *** | (0.036) ** | (0.041) ** |
| | Periphery | -0.037 | (0.016) ** | (0.021) * | (0.020) * |
| | Core − Periphery | 0.127 | (0.036) *** | (0.044) *** | (0.047) *** |
| Sales | Core | 0.063 | (0.034) * | (0.028) ** | (0.031) ** |
| | Periphery | -0.078 | (0.025) *** | (0.041) * | (0.047) * |
| | Core − Periphery | 0.142 | (0.042) *** | (0.047) *** | (0.056) ** |
| Exports | Core | 0.219 | (0.050) *** | (0.062) *** | (0.063) *** |
| | Periphery | -0.160 | (0.056) *** | (0.073) ** | (0.065) ** |
| | Core − Periphery | 0.379 | (0.075) *** | (0.086) *** | (0.061) *** |

*Notes:* This table reports the split-sample PPML estimates from Table 2 under alternative inference procedures. Columns (1) and (2) reproduce the coefficient estimates and standard errors clustered at the county level. Column (3) reports exposure-robust standard errors treating the 425 four-digit CIC industries as independent shocks, following Adao et al. (2019). Column (4) allows the four-digit industry shocks to be arbitrarily correlated within 29 two-digit industries. For the core–periphery difference, columns (3) and (4) use the stacked sample to account for the fact that the two groups are exposed to the same industry shocks. Stars indicate statistical significance under the corresponding inference procedure. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A15: Pareto Shape Estimates by Period: All Active Firms

| | (1) All | (2) Top 50% | (3) Top 25% | (4) Top 10% | (5) Top 5% | (6) Top 3% | (7) Top 1% |
|---|---|---|---|---|---|---|---|
| **Panel A. 1998–2000** | | | | | | | |
| $\alpha_C$ | 2.717 | 3.223 | 3.508 | 3.730 | 3.837 | 3.905 | 4.099 |
| | (0.009) | (0.017) | (0.027) | (0.044) | (0.062) | (0.079) | (0.123) |
| $\alpha_P$ | 3.182 | 3.871 | 4.307 | 4.722 | 4.954 | 5.138 | 5.740 |
| | (0.018) | (0.036) | (0.060) | (0.106) | (0.149) | (0.192) | (0.337) |
| $\alpha_C - \alpha_P$ | -0.465*** | -0.648*** | -0.799*** | -0.992*** | -1.117*** | -1.233*** | -1.641*** |
| | (0.020) | (0.040) | (0.066) | (0.115) | (0.162) | (0.207) | (0.359) |
| # of Obs. (Core) | 262,909 | 131,455 | 65,727 | 26,291 | 13,145 | 7,887 | 2,629 |
| # of Obs. (Periphery) | 77,029 | 38,515 | 19,257 | 7,703 | 3,851 | 2,311 | 770 |
| **Panel B. 2001–2003** | | | | | | | |
| $\alpha_C$ | 2.613 | 3.084 | 3.316 | 3.488 | 3.584 | 3.656 | 3.962 |
| | (0.008) | (0.015) | (0.024) | (0.039) | (0.054) | (0.066) | (0.108) |
| $\alpha_P$ | 3.014 | 3.715 | 4.116 | 4.422 | 4.635 | 4.792 | 5.123 |
| | (0.015) | (0.032) | (0.055) | (0.092) | (0.135) | (0.177) | (0.281) |
| $\alpha_C - \alpha_P$ | -0.401*** | -0.631*** | -0.800*** | -0.934*** | -1.051*** | -1.136*** | -1.161*** |
| | (0.017) | (0.036) | (0.060) | (0.100) | (0.145) | (0.189) | (0.301) |
| # of Obs. (Core) | 335,694 | 167,847 | 83,924 | 33,569 | 16,785 | 10,071 | 3,357 |
| # of Obs. (Periphery) | 89,557 | 44,779 | 22,389 | 8,956 | 4,478 | 2,687 | 896 |
| **Panel C. 2005–2007** | | | | | | | |
| $\alpha_C$ | 2.453 | 2.965 | 3.206 | 3.375 | 3.456 | 3.501 | 3.715 |
| | (0.006) | (0.011) | (0.017) | (0.028) | (0.037) | (0.046) | (0.074) |
| $\alpha_P$ | 2.691 | 3.496 | 3.853 | 4.080 | 4.253 | 4.363 | 4.603 |
| | (0.010) | (0.022) | (0.037) | (0.062) | (0.088) | (0.113) | (0.185) |
| $\alpha_C - \alpha_P$ | -0.237*** | -0.531*** | -0.647*** | -0.705*** | -0.797*** | -0.862*** | -0.889*** |
| | (0.011) | (0.025) | (0.041) | (0.068) | (0.096) | (0.122) | (0.199) |
| # of Obs. (Core) | 625,853 | 312,927 | 156,463 | 62,585 | 31,293 | 18,776 | 6,259 |
| # of Obs. (Periphery) | 182,537 | 91,269 | 45,634 | 18,254 | 9,127 | 5,476 | 1,825 |

*Notes:* This table reports Pareto shape parameters estimated from QQ regressions on firm sales, separately for core and periphery counties within each period. Panel A reproduces Table 6 and is repeated here so that the three windows can be read side by side. "Top X%" restricts the sample to firms above the corresponding percentile of the sales distribution within each group. Shape parameters are obtained by mapping the estimated QQ slopes under a trade elasticity of $\sigma = 4$. Standard errors are obtained from 500 firm-level block bootstrap replications, resampling firms with replacement and retaining all yearly observations for each sampled firm. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A16: Pareto Shape Estimates by Period: Entering Firms

| | (1) All | (2) Top 50% | (3) Top 25% | (4) Top 10% | (5) Top 5% | (6) Top 3% | (7) Top 1% |
|---|---|---|---|---|---|---|---|
| **Panel A. 1999–2000** | | | | | | | |
| $\alpha_C$ | 3.211 | 3.598 | 3.830 | 3.896 | 3.849 | 3.745 | 3.270 |
| | (0.021) | (0.038) | (0.066) | (0.111) | (0.155) | (0.188) | (0.223) |
| $\alpha_P$ | 3.828 | 4.264 | 4.564 | 4.931 | 5.177 | 5.455 | 5.846 |
| | (0.045) | (0.072) | (0.109) | (0.188) | (0.277) | (0.393) | (0.761) |
| $\alpha_C - \alpha_P$ | -0.616*** | -0.666*** | -0.734*** | -1.036*** | -1.328*** | -1.710*** | -2.576*** |
| | (0.049) | (0.081) | (0.127) | (0.218) | (0.317) | (0.436) | (0.793) |
| # of Obs. (Core) | 31,871 | 15,936 | 7,968 | 3,187 | 1,594 | 956 | 319 |
| # of Obs. (Periphery) | 10,164 | 5,082 | 2,541 | 1,016 | 508 | 305 | 102 |
| **Panel B. 2001–2003** | | | | | | | |
| $\alpha_C$ | 3.158 | 3.507 | 3.689 | 3.761 | 3.755 | 3.748 | 3.690 |
| | (0.014) | (0.024) | (0.037) | (0.062) | (0.088) | (0.113) | (0.175) |
| $\alpha_P$ | 3.475 | 3.985 | 4.352 | 4.668 | 4.787 | 4.840 | 4.820 |
| | (0.024) | (0.042) | (0.070) | (0.124) | (0.182) | (0.236) | (0.377) |
| $\alpha_C - \alpha_P$ | -0.316*** | -0.479*** | -0.663*** | -0.907*** | -1.032*** | -1.092*** | -1.131*** |
| | (0.027) | (0.049) | (0.080) | (0.139) | (0.202) | (0.262) | (0.416) |
| # of Obs. (Core) | 79,386 | 39,693 | 19,847 | 7,939 | 3,969 | 2,382 | 794 |
| # of Obs. (Periphery) | 23,376 | 11,688 | 5,844 | 2,338 | 1,169 | 701 | 234 |
| **Panel C. 2005–2007** | | | | | | | |
| $\alpha_C$ | 3.006 | 3.319 | 3.504 | 3.582 | 3.616 | 3.640 | 3.759 |
| | (0.011) | (0.019) | (0.030) | (0.046) | (0.064) | (0.080) | (0.124) |
| $\alpha_P$ | 3.196 | 3.707 | 4.143 | 4.454 | 4.457 | 4.436 | 4.552 |
| | (0.016) | (0.031) | (0.055) | (0.102) | (0.143) | (0.182) | (0.348) |
| $\alpha_C - \alpha_P$ | -0.190*** | -0.388*** | -0.638*** | -0.872*** | -0.841*** | -0.796*** | -0.793** |
| | (0.020) | (0.036) | (0.062) | (0.112) | (0.156) | (0.199) | (0.369) |
| # of Obs. (Core) | 99,496 | 49,748 | 24,874 | 9,950 | 4,975 | 2,985 | 995 |
| # of Obs. (Periphery) | 40,424 | 20,212 | 10,106 | 4,042 | 2,021 | 1,213 | 404 |

*Notes:* This table reports Pareto shape parameters estimated from QQ regressions on firm sales, separately for core and periphery counties within each period and restricted to firms absent from the panel in the preceding year. Because no preceding year is observed for 1998, the first period covers 1999–2000. "Top X%" restricts the sample to firms above the corresponding percentile of the sales distribution within each group. Shape parameters are obtained by mapping the estimated QQ slopes under a trade elasticity of $\sigma = 4$. Standard errors are obtained from 500 firm-level block bootstrap replications, resampling firms with replacement and retaining all yearly observations for each sampled firm. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A17: Industrial Concentration versus Population Agglomeration

| | (1) Firms | (2) Employment | (3) Sales | (4) Exports |
|---|---|---|---|---|
| **Quartiles of Manufacturing Employment Share** | | | | |
| **Panel A. Core** | | | | |
| NTR gap × Post | 0.0865** | 0.0534* | 0.00830 | 0.155*** |
| | (0.0336) | (0.0275) | (0.0310) | (0.0533) |
| NTR gap × Density × Post | 0.0743*** | 0.0874*** | 0.0952*** | 0.0818** |
| | (0.0197) | (0.0192) | (0.0257) | (0.0388) |
| # of Obs. | 5,220 | 5,220 | 5,220 | 5,170 |
| **Panel B. Periphery** | | | | |
| NTR gap × Post | -0.0177 | -0.0359** | -0.0478* | -0.0960* |
| | (0.0154) | (0.0164) | (0.0244) | (0.0579) |
| NTR gap × Density × Post | -0.0158 | -0.00273 | -0.0844*** | -0.141** |
| | (0.0179) | (0.0184) | (0.0291) | (0.0567) |
| # of Obs. | 15,150 | 15,150 | 15,150 | 12,050 |
| **Panel C. All Counties** | | | | |
| NTR gap × Post | -0.0211 | -0.0604*** | -0.113*** | -0.0805* |
| | (0.0218) | (0.0182) | (0.0275) | (0.0478) |
| NTR gap × Density × Post | -0.0168 | -0.0214 | -0.0481 | -0.136** |
| | (0.0251) | (0.0230) | (0.0315) | (0.0545) |
| NTR gap × Mfg. Share × Post | 0.0560*** | 0.0636*** | 0.0790*** | 0.105*** |
| | (0.0126) | (0.0122) | (0.0114) | (0.0137) |
| # of Obs. | 20,370 | 20,370 | 20,370 | 17,220 |
| County FE | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ |
| Tariff Controls | ✓ | ✓ | ✓ | ✓ |
| Trends | ✓ | ✓ | ✓ | ✓ |

*Notes:* This table compares initial industrial concentration and population agglomeration as moderators of county exposure to TPU reduction. Panels A and B report separate estimates for core and periphery counties, respectively, with pre-treatment population density as the moderator. Panel C pools all counties and includes interactions with both pre-treatment population density and manufacturing employment share, standardized to have mean zero and unit variance. All specifications are estimated using PPML with the fixed effects, tariff controls, and trends in Table 2. Standard errors are clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A18: Parameter Values

| Parameter | Symbol | Value | Source |
|---|---|---|---|
| **Estimated** | | | |
| Pareto shape, Core | $\alpha_C$ | 3.730 | Own estimation (Table 6) |
| Pareto shape, Periphery | $\alpha_P$ | 4.722 | Own estimation (Table 6) |
| **From literature** | | | |
| Elasticity of substitution | $\sigma$ | 4 | Broda and Weinstein (2006) |
| Agricultural labor share | $\eta$ | 0.5 | Restuccia et al. (2008) |
| Migration elasticity | $\nu$ | 5 | See text |
| **Calibrated** | | | |
| Agricultural TFP | $a_A$ | 1 | Normalized |
| Subsistence requirement | $\bar{A}$ | 1 | Normalized |
| Fixed costs | $(f^e, f^d, f^x)$ | (2, 3, 9) | $f^e < f^d < f^x$; see text |
| Land endowment | $(\bar{T}_C, \bar{T}_P)$ | (30, 170) | Core share of agricultural land |
| Foreign demand | $\Phi_F$ | 62.4 | Match $L_C/L_P$ |

*Notes:* This table reports the parameter values used in the baseline simulation. The migration elasticity $\nu = 5$ is only used in solving the model across a range of variable trade costs $\tau_F$ to illustrate the mechanisms. Results are also reported for a range of $\nu$, identifying a threshold $\nu^*$ above which manufacturing employment in the periphery declines after trade liberalization. $\Phi_F \equiv P_F^{\sigma-1} E_F$ is a composite parameter, as the foreign price index and expenditure are not separately identified.

# B Theory Appendix

This appendix provides supplementary material for the theoretical framework developed in Section 7. It proves Proposition 1, which decomposes the response of manufacturing employment into wage and migration channels, and derives closed-form expressions for the equilibrium integrals under the Pareto productivity assumption.

## B.1 Proofs

**Proof of Proposition 1.** Starting from the regional labor market clearing condition $L_i = L_i^M + L_i^A$, the manufacturing share is:

$$s_i = \frac{L_i - L_i^A}{L_i} = 1 - \frac{L_i^A}{L_i}. \tag{B.1}$$

Taking the derivative with respect to $\tau$:

$$\frac{\partial s_i}{\partial \tau} = -\frac{\partial (L_i^A / L_i)}{\partial \tau} = -\frac{L_i^A}{L_i}\left[\frac{\partial \ln L_i^A}{\partial \tau} - \frac{\partial \ln L_i}{\partial \tau}\right]. \tag{B.2}$$

Since $(1-s_i) = L_i^A / L_i$, and inverting the agricultural first-order condition gives $L_i^A = \bar{T}_i \left(p_A \eta a_A / w_i\right)^b$, where $b \equiv 1/(1-\eta)$, we have

$$\frac{\partial \ln L_i^A}{\partial \tau} = -b \, \frac{\partial \ln w_i}{\partial \tau}. \tag{B.3}$$

Substitution yields:

$$\frac{\partial s_i}{\partial \tau} = (1-s_i)\left[b \, \frac{\partial \ln w_i}{\partial \tau} + \frac{\partial \ln L_i}{\partial \tau}\right]. \tag{B.4}$$

Turning to manufacturing employment, $L_i^M = s_i L_i$ implies:

$$\frac{d\ln L_i^M}{d\tau} = \frac{1}{s_i}\frac{ds_i}{d\tau} + \frac{d\ln L_i}{d\tau} = \frac{1-s_i}{s_i}\cdot b \cdot \frac{d\ln w_i}{d\tau} + \frac{1}{s_i}\cdot\frac{d\ln L_i}{d\tau}, \tag{B.5}$$

which is the desired decomposition into the wage effect and the migration effect. ■

## B.2 Closed-Form Expressions under Pareto Productivity

Under the Pareto assumption $G_i(\varphi) = 1-\varphi^{-\alpha_i}$ with $\alpha_i > \sigma - 1$, the conditional expectation of $(\varphi/\varphi_i^d)^{\sigma-1}$ over active firms ($\varphi \geq \varphi_i^d$) reduces to a constant that depends only on the shape parameter and the elasticity of substitution:

$$\mathbb{E}\left[\left(\frac{\varphi}{\varphi_i^d}\right)^{\sigma-1}\middle|\, \varphi \geq \varphi_i^d\right] = \frac{\alpha_i}{\alpha_i - (\sigma-1)} \equiv \kappa_i > 1. \tag{B.6}$$

This ratio governs the relationship between average and cutoff-level revenues, and appears throughout the expressions below.

**Free entry condition (Equation 17).** The expected profit of an entrant in region $i$ is:

$$\int_{\varphi_i^d}^{\infty} \pi_i(\varphi)\, dG_i(\varphi) = (\varphi_i^d)^{-\alpha_i}(\kappa_i - 1)\, w_i \left[f^d + \left(\frac{\varphi_i^d}{\varphi_i^x}\right)^{\alpha_i} f^x\right], \tag{B.7}$$

so the free entry condition (17) becomes:

$$f^e = (\varphi_i^d)^{-\alpha_i}(\kappa_i - 1)\left[f^d + \left(\frac{\varphi_i^d}{\varphi_i^x}\right)^{\alpha_i} f^x\right]. \tag{B.8}$$

**Price index (Equation 18).** The contribution of region $i$'s active firms to the common price index is:

$$M_i \int_{\varphi_i^d}^{\infty} p_i(\varphi)^{1-\sigma} \frac{dG_i(\varphi)}{1-G_i(\varphi_i^d)} = M_i \left(\frac{\sigma}{\sigma-1} w_i\right)^{1-\sigma} (\varphi_i^d)^{\sigma-1} \cdot \kappa_i. \tag{B.9}$$

Therefore:

$$(P^M)^{1-\sigma} = \sum_{j \in \{C,P\}} M_j \left(\frac{\sigma}{\sigma-1} w_j\right)^{1-\sigma} (\varphi_j^d)^{\sigma-1} \cdot \kappa_j. \tag{B.10}$$

**Manufacturing labor demand (Equation 21).** The variable labor input of a firm with productivity $\varphi$ serving market $j$ is

$$\ell_{ij}(\varphi) = \frac{\sigma-1}{\sigma} \cdot \frac{r_{ij}(\varphi)}{w_i}. \tag{B.11}$$

To obtain a closed-form expression, I substitute out revenue using the zero-profit conditions. Let $r_{iH} \equiv r_{iC} + r_{iP}$ denote combined domestic revenue. The domestic zero-profit condition implies

$$r_{iH}(\varphi_i^d) = \sigma w_i f^d. \tag{B.12}$$

Since CES demand yields $r_{iH}(\varphi) = r_{iH}(\varphi_i^d) \cdot (\varphi/\varphi_i^d)^{\sigma-1}$ for any active firm, substitution gives:

$$\ell_i^H(\varphi) = \frac{\sigma-1}{\sigma} \cdot \frac{\sigma w_i f^d}{w_i} \left(\frac{\varphi}{\varphi_i^d}\right)^{\sigma-1} = (\sigma-1) f^d \left(\frac{\varphi}{\varphi_i^d}\right)^{\sigma-1}, \tag{B.13}$$

where the $1/\sigma$ from the variable cost share is exactly offset by the $\sigma$ in the zero-profit condition. Taking the conditional expectation over active firms under the Pareto distribution:

$$\mathbb{E}\left[\ell_i^H(\varphi) \mid \varphi \geq \varphi_i^d\right] = (\sigma-1)\kappa_i f^d. \tag{B.14}$$

By identical logic, the export zero-profit condition $r_{iF}(\varphi_i^x) = \sigma w_i f^x$ implies that the average variable export labor per exporting firm is $(\sigma-1)\kappa_i f^x$. Weighting by the exporter share $p_i^x \equiv (\varphi_i^d/\varphi_i^x)^{\alpha_i}$ among active firms, the total average operating labor per active firm—combining variable and fixed costs for both domestic and export activities—is

$$\bar{\ell}_i^{\text{op}} = [(\sigma-1)\kappa_i + 1]\left(f^d + p_i^x f^x\right). \tag{B.15}$$

Manufacturing labor demand in region $i$ therefore equals

$$L_i^M = M_i\left[\left(\frac{\varphi_i^d}{\underline{\varphi}}\right)^{\alpha_i} f^e + [(\sigma-1)\kappa_i + 1]\left(f^d + p_i^x f^x\right)\right], \tag{B.16}$$

where the first term accounts for entrants' sunk costs, recalling that $(\varphi_i^d/\underline{\varphi})^{\alpha_i} = M_i^e/M_i$ is the inverse survival rate.

# C Build a Time-consistent List of Chinese Counties

This section describes the construction of a time-consistent list of Chinese counties used in the main analysis. Counties form the third tier of China's administrative hierarchy, below provinces and prefecture-level cities. At this level, there are eight official administrative types, though three account for the vast majority of units: (normal) counties, county-level cities, and urban districts.[1] To harmonize all these county-level administrative units while preserving the continuity of metropolitan areas, I aggregate all urban districts within each prefecture-level city as of 2010 into a single hypothetical "county" representing the metropolitan area, and treat all remaining non-district units as separate counties.

## C.1 General Principles

Chinese prefecture-level cities function as regional administrative governments overseeing multiple county-level units. A typical prefecture-level city governs a mix of urban districts (one of which hosts the municipal government), counties, and county-level cities. Despite this administrative division, the metropolitan area usually consists of a contiguous urban core formed by one or more urban districts, surrounded by subordinate counties and county-level cities. As urban areas expand, nearby subordinate units are sometimes incorporated and reclassified as urban districts. While the urban core exhibits dense commuting flows, unincorporated counties and county-level cities tend to retain more distinct local labor markets.

The objective is to identify county-level administrative units whose boundaries remain stable over the sample period from 1998 to 2007. Because the analysis is conducted at the county

[1] In 2000, the breakdown is as follows: counties (1,503), county-level cities (400), urban districts (787), autonomous counties (116), banners (49), autonomous banners (3), special districts (2), and forestry districts (1).

level, most prefecture-level adjustments (such as reassigning a county from one prefecture-level city to another) can be safely ignored. Changes that do not alter county boundaries, including "upgrading" a county to a county-level city, are likewise irrelevant. The only adjustments that matter are those that change the effective county-level jurisdiction, which in practice arise from two types of changes involving urban districts.

First, municipal governments frequently merge, split, or reassign urban districts. Because these districts jointly constitute the city's metropolitan area, such changes typically reflect the evolution of urban morphology and the reorganization of functional zones rather than economically meaningful boundary changes at the county level. For the purposes of this study, I therefore aggregate all urban districts within a prefecture-level city into a single hypothetical "county." This aggregation abstracts from administrative reshuffling within the urban core and allows the analysis to focus on the metropolitan area as a unified economic unit. An additional advantage of this approach is that the merged urban districts generally exhibit stable and well-documented boundaries over time, which further improves temporal consistency.

Second, as metropolitan areas expand, surrounding counties or county-level cities may be incorporated and reclassified as urban districts. These incorporations, commonly referred to as *chexian shequ* (literally, "dissolving a county to establish an urban district"), became increasingly common in the 2010s. To limit boundary inconsistencies, I apply a timing-based rule: only counties and county-level cities incorporated into urban districts before 2010 are treated as part of the metropolitan area and merged into the hypothetical "county." Units incorporated after 2010 are instead retained as independent county-level observations.

Together, these principles yield a time-consistent list of county-level administrative units.[2] All units are uniformly referred to as "counties." I further classify each unit by administrative status into three categories: city (merged urban districts), county-level city, and county. The "county" category includes standard counties as well as special county-level units, including autonomous counties, banners, autonomous banners, special districts, and forestry districts.

## C.2 Adjustments to Direct-administered Municipalities

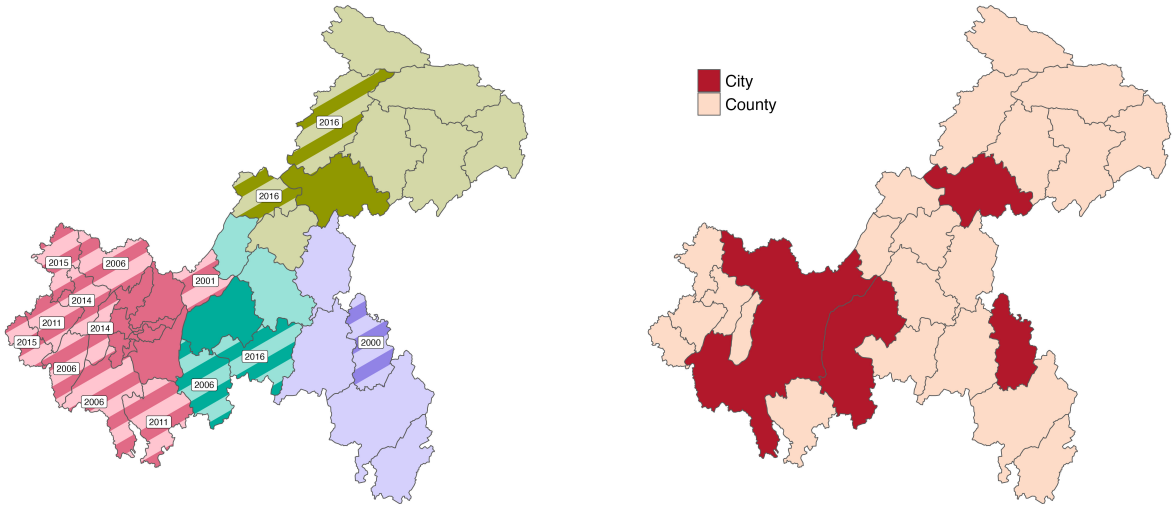
China has four direct-administered municipalities (*zhixiashi*)—Beijing, Tianjin, Shanghai, and Chongqing—which report directly to the central government rather than to any province. By 2016, all county-level units in Beijing, Tianjin, and Shanghai had been converted into urban districts. I therefore merge all constituent units within each of these municipalities and treat the municipality as a single county-level unit. This construction, however, implies a "province" containing only one county-level unit, which makes province-year fixed effects ill-defined. To address this issue, I assign each municipality to a neighboring province: Beijing and Tianjin to Hebei Province, and Shanghai to Jiangsu Province.

Chongqing requires a different treatment. Although it is also directly administered, it is comparable in scale to a small province and consists of four former prefecture-level cities—Chongqing, Wanxian, Fuling, and Qianjiang—that were separated from Sichuan Province in 1997. Instead of merging all constituent units, I treat Chongqing as a hypothetical province composed of these four prefecture-level cities and apply the same aggregation rules to their subordinate counties. The next subsection illustrates my approach using this special case as an example.

[2]Other boundary changes, such as county splits or mergers, are rare. In such cases, I aggregate the affected units and use the resulting time-consistent geographic unit throughout the sample period.

## C.3 An Example

Chongqing provides a useful illustration as it combines (i) a directly administered municipality, (ii) multiple prefecture-level jurisdictions, and (iii) substantial district–county reclassification during the 2000s. Figure C1 visualizes both the raw administrative geography and the harmonized units used in the analysis. The left panel plots Chongqing's four (hypothetical) prefecture-level cities. Urban districts already in place by 2000 are shaded in darker colors to indicate the pre-existing urban core. Counties that were subsequently incorporated into the urban core and reclassified as urban districts are marked with stripes, and the year of incorporation is labeled for each such unit. This panel thus distinguishes stable urban districts from county-to-district conversions that would otherwise generate boundary breaks over time. The right panel applies the harmonization rules. Within each prefecture, all urban districts observed by 2010 are aggregated into a single hypothetical "county" representing the metropolitan area of that prefecture. County-level units that were not incorporated by 2010 are kept as separate "counties," preserving their original jurisdictions. As Chongqing currently has no county-level cities, all remaining non-district units in the figure are normal counties.




(a) Administrative Divisions (b) Time-consistent Counties

Figure C1: Constructing Time-Consistent County Units in Chongqing

*Notes:* This figure shows the construction of time-consistent county-level units in Chongqing. Panel (a) shows the raw administrative divisions, with urban districts shaded in darker colors and counties later incorporated and reclassified as urban districts marked with stripes; the year of incorporation is labeled for each unit. Panel (b) applies the harmonization rules by aggregating all urban districts within each former prefecture-level city as of 2010 into a single hypothetical county, while keeping all remaining counties as separate units.

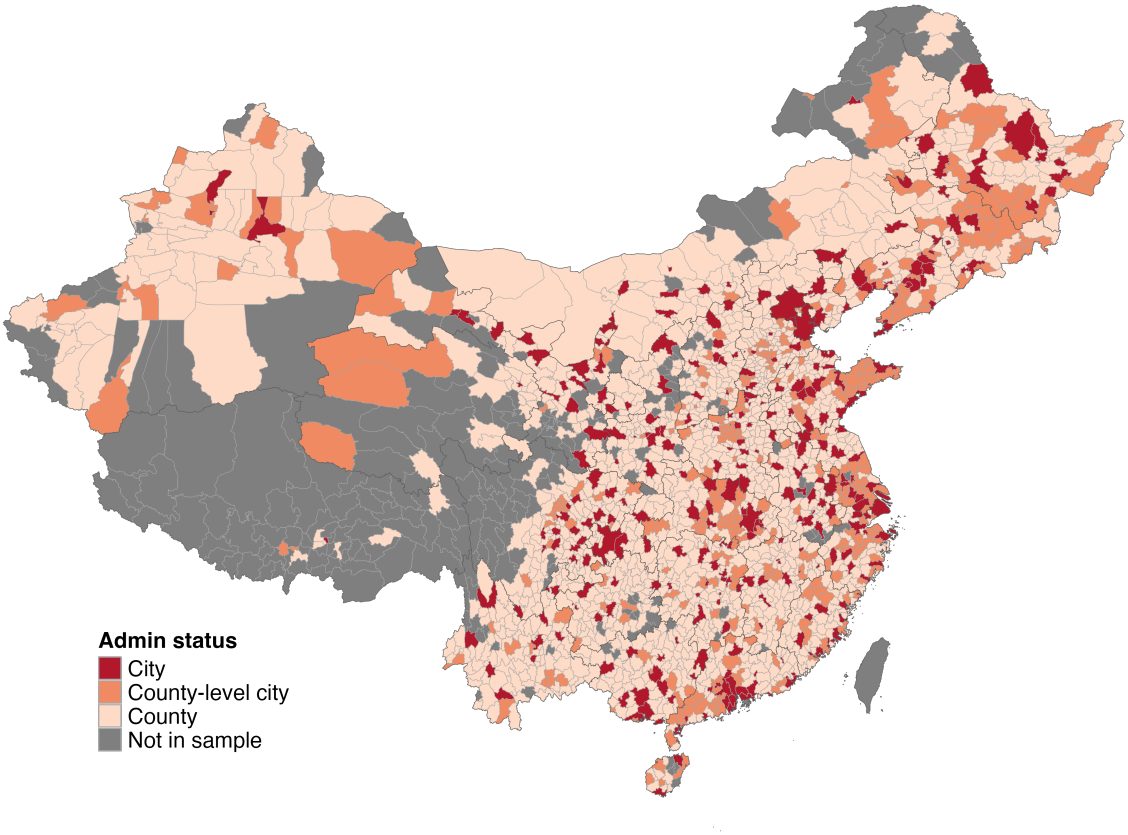



Figure C2: Geographic Distribution of Time-Consistent County Units

*Notes:* This figure maps the time-consistent county-level administrative units constructed for the main analysis. Units are classified by administrative status as of 2010, distinguishing urban districts (city), county-level cities, and counties. Counties outside the estimation sample because an NTR gap cannot be constructed are shaded in gray.